

Highlights

Sensor noise, array design and helmet fit in on-scalp and cryogenic magnetoencephalography: a simulation study from adults to children

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- In simulations, OPM sensor noise or array design could reverse which system led
- Measured SQUID noise shifted detectability as much as the modeled OPM advantage
- A helmet fitted to the adult's gap removed most of the smaller heads' extra gain
- With an idealized detector, a dense OPM array detected weaker shallow spikes
- Exploratory: OPMs at the SQUID sensor sites trailed on ventral and polar cortex

Sensor noise, array design and helmet fit in on-scalp and cryogenic magnetoencephalography: a simulation study from adults to children

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Abstract

On-scalp optically pumped magnetometers (OPMs) record larger fields than cryogenic magnetoencephalography (MEG), but in a spherical head their signal-to-noise ratio is higher only for sources shallower than a depth set by the sensor-noise ratio. Here, with explicit sensor, brain and room noise, we compared the 306-channel Neuromag system with a dense and a site-matched OPM array (at Neuromag's sites) by matched-filter detectability in an adult and eight smaller heads (two scaled adults, three infant templates, three children). In the adult, with sensor plus brain noise, the dense 15-fT/ $\sqrt{\text{Hz}}$ array had 1.14 [1.12, 1.17] times Neuromag's detectability (interval: spatial variation), an advantage not established at 30 fT/ $\sqrt{\text{Hz}}$; the site-matched array showed no established advantage. Measured noise raised Neuromag's detectability 1.14-fold over the model, so the dense advantage awaits measured OPM noise. Under the modeled noise (exploratory), the dense advantage was smallest in cingulate cortex, and the site-matched array trailed on ventral and polar surfaces. In the fixed adult helmet, the scaled adults and templates gained +0.44 to +1.07 dB over the adult, and +0.10 to +0.23 dB with each head's helmet fitted at the adult's top-contact gap; the children, with unverified near-scalp anatomy, are provisional. In a confirmatory simulation (idealized detector, modeled noise, fixed adult helmet), the dense array detected spikes 10–20 mm deep at 21–32 % lower strength than Neuromag in all nine anatomies (not independent replications); exploratory localization established no difference. Clinical reports of higher OPM signal-to-noise ratios are compatible with, but not explained by, these conditions.

Keywords: Magnetoencephalography (MEG), Optically pumped magnetometer (OPM), Superconducting quantum interference device (SQUID), Pediatric neuroimaging, Helmet fit, Noise covariance, Interictal spikes

1. Introduction

Optically pumped magnetometers (OPMs) measure the brain's magnetic field at room temperature and can be worn on the scalp, whereas the superconducting quantum interference devices (SQUIDs) of conventional magnetoencephalography (MEG) sit in a rigid cryogenic helmet (Boto et al., 2017; Iivanainen et al., 2017). The helmet is built for adults, so a child's head sits in it with a wide and uneven gap, whereas an on-scalp array can fit any head (Feys and De Tiège, 2024; Hill et al., 2019). Clinical OPM studies of epilepsy report larger interictal discharges than cryogenic

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MEG (Feys et al., 2022). Their signal-to-noise ratios (SNRs) were higher in some comparisons (Feys et al., 2022, 2025; Ren et al., 2025, abstract only) and equal or slightly lower in others (Hillebrand et al., 2023; Schwartz et al., 2025), and their weakest results concern temporal and mesial temporal sources (Feys et al., 2024, 2025; Shen et al., 2026).

A larger field does not by itself give a higher SNR, because an OPM is a noisier sensor than a SQUID magnetometer. Jas et al. (2026) showed this in a spherical head with a point magnetometer on the scalp (OPM) and one 18 mm above it (SQUID). If the OPM’s noise is η times the SQUID’s, its single-channel SNR is higher only for sources shallower than an equal-SNR depth, about 28 mm at $\eta = 3$ in an adult head. With a SQUID shell that follows the head at a constant standoff, this depth covers a larger fraction of a small brain, which predicts a larger OPM benefit in children.

A realistic comparison needs explicit noise seen through each array’s own lead fields, finite arrays on a real scalp, and a real helmet around a smaller head. With sensor noise only, an array of helium-4 OPMs gained on SQUID magnetometers as the head shrank, both in a standard adult placement and in a helmet fitted at a constant gap (Zahran et al., 2022). With brain noise in one adult, simulated spikes were visible on the scalp from smaller cortical patches with on-scalp than with conventional MEG (Westin et al., 2023). With brain noise but no sensor noise, OPMs placed at the SQUID sites of epilepsy patients raised a channel-averaged SNR in no region, although the OPM–SQUID difference followed the ratio of sensor distances, which tracked head circumference (Matsubara et al., 2026).

These studies leave three gaps. None compared head size and helmet fit with brain noise included, and none checked its noise model against a measured covariance. The one helmet fitted at a constant gap was fitted at the adult’s closest distance (Zahran et al., 2022), not at the adult’s median gap. It is therefore unknown whether the larger OPM advantage expected in children reflects the small head itself or the poor fit of the adult helmet. It is also unknown how far a modeled comparison can be trusted while the OPM’s noise has not been measured in the same conditions.

Here, we compared the 306-channel Neuromag system with a dense and a site-matched OPM array in a realistic adult head and eight smaller heads, using known-topography detectability with explicit sensor, brain and room noise. We asked four questions. First, what decides the comparison in the adult, and how far can the noise model be trusted when it is checked against Neuromag’s measured noise? Second, which brain regions favor the OPM, and where does an array at Neuromag’s sites fall behind (exploratory)? Third, is the larger relative advantage in smaller heads a matter of head size or of the fixed adult helmet? We tested this with a helmet fitted to each head at the adult’s gap. Fourth, does the detectability difference change which simulated interictal spikes are detected, in a confirmatory test, and how well are they localized (exploratory)?

2. Materials and methods

2.1. Study design and estimands

We compared arrays by known-topography detectability, the output SNR of the matched filter for a source of known field pattern in noise of known covariance:

$$d = \sqrt{\mathbf{s}^\top \mathbf{C}^{-1} \mathbf{s}}, \quad (1)$$

where $\mathbf{s}$ is the noise-free field of a 10-nAm cortical dipole at the array’s channels and $\mathbf{C}$ is the array’s noise covariance (Section 2.4), taken as known (the oracle covariance). Two arrays were

compared target by target by the paired ratio $d_{\text{OPM}}/d_{\text{Neuromag}}$ or, in decibels, by

$$D = 20 \log_{10} \frac{d_{\text{OPM}}}{d_{\text{Neuromag}}}, \quad (2)$$

where D is positive when the OPM array detects the source better. Neuromag denotes all 306 channels unless stated otherwise.

For the adult, we report the unweighted median of the paired ratio over all 7,661 targets, including the 558 on the medial wall, which is not cortex; this is the adult analysis’s declared convention, and without them the dense array’s ratio is 1.15. The smaller heads differ in cortical area, so for them we report D as an area-weighted median over cortical targets, medial wall excluded, and its change from the adult, $\Delta = D_{\text{head}} - D_{\text{adult}}$, with the adult placed by the same helmet rule. The other estimands are introduced below and summarized in Section 2.6.

2.2. Head models

The adult head was the MNE sample subject (Gramfort et al., 2013), with FreeSurfer surfaces and atlas labels (Desikan et al., 2006; Fischl, 2012). Targets were the vertices of an MNE-Python oct-6 source space (a recursively subdivided octahedron) on the white-matter surface lying at least 4 mm inside the inner skull, each a 10-nAm dipole normal to the cortex. A target’s depth was its distance to the nearest point of the MRI scalp. A three-layer boundary-element model (BEM; Hämäläinen and Sarvas, 1989) with conductivities of 0.3, 0.006, 0.3 S/m had its outer surface on the MRI scalp.

The eight smaller heads fell into three classes (Table 1). The two scaled adults were copies of the adult scaled to school-age size and to the head circumference of the 24-month template. The three infant templates, at 24, 18 and 12 months, were averages of many MRIs (O’Reilly et al., 2021; Richards et al., 2016). The three typically developing school-aged children, A, B and C (7.8–8.7 years), came from OpenNeuro dataset ds005234 (Fadeev et al., 2024, 2025). The nine anatomies are therefore not independent anatomical replications, and all of them share the noise calibration, the adult’s conductivities and the processing, so agreement across them does not test those assumptions.

The children’s anatomy needed three further steps (Section S4). First, the dataset’s BEM surfaces placed the inner skull a median of 0.8–2.5 mm below the scalp over the upper head, implausibly close. We moved it inward toward 8 mm below the scalp wherever it lay shallower, keeping every inner-skull vertex at least 2 mm from the white-surface vertices, so that over the upper head it lay a median of 5.7–7.8 mm below the scalp (adult, 9.7 mm). Between vertices, the inner skull came within 0.9–1.3 mm of the white surface, and in child B the outer skull came within 0.17 mm of the scalp. A single-compartment BEM, which needs no skull, changed the difference between each smaller head’s median D and the adult’s by -0.06 to $+0.01$ dB.

Second, no fiducials come with these data. We transferred them from the adult by a similarity transform fitted between the adult’s and the child’s white surfaces (trimmed iterative closest points) and moved each to the nearest scalp vertex. Applied to the infant templates, this method reproduced their own fiducials with errors of 1.6–14.4 mm and their head frames with rotations of 3.2 – 5.9° . Third, because the children’s white surfaces came within 3.7–5.7 mm of the scalp, we checked the surfaces against their T1-weighted MRIs (Fig. S6). The FreeSurfer scalp lay about as far inside the T1 head boundary in the children (median 1.3–1.5 mm) as in the adult (1.3 mm), and the white surfaces were registered to the T1 images. Relative to these images, the shallow cortex is therefore not a scalp-placement or registration error.

The count of shallow targets depends on the scalp definition: measured to the MRI head boundary, child B has 47 rather than 247 targets less than 10 mm deep. The source rule left out

Table 1. Head models.

Head	Class	Head circumference (cm)	Cortex (cm ²)	Targets	OPM sites, dense / site-matched
Adult	MNE sample subject	58.6	1,878	7,103	208 / 98
School-age size	scaled adult	52.5	1,470	6,949	174 / 89
2-year size	scaled adult	49.5	1,290	6,855	155 / 90
24-month template	infant template	49.5	1,062	7,559	151 / 83
18-month template ^a	infant template	49.1	975	7,363	157 / 80
12-month template	infant template	46.9	895	7,592	144 / 82
Child A (7.8 years)	school-aged child	52.0	1,852	7,138	155 / 90
Child B (8.3 years)	school-aged child	48.6	1,666	6,866	153 / 85
Child C (8.7 years)	school-aged child	53.5	1,869	7,392	167 / 91

Head circumference is that of the MRI scalp; cortex is the usable cortical area; targets exclude the medial wall. The median magnetometer-to-scalp gaps of each head in each helmet, with the fitted helmets' scale factors, are listed in Table S5. Scaled adults: copies of the adult (Section 2.2). Infant templates: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016). Children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025); provisional examples (Section 2.2). ^aThe MRI check classes this template as misregistered: its white surface fits its T1 image best after a 2.3-mm rigid shift (Section 2.2).

3.3–10.0 % of the children's cortex (adult, 7.2 %), including 94–100 % of their cortex within 8 mm of the scalp. Because the MRI check cannot verify the cortex nearest the scalp (Section 4.4), we treat the children's results as provisional geometric examples. The infant templates are averages. The same check placed their scalps 1.3–1.6 mm outside their own MRI head boundaries, whereas the adult's lies inside. The check also classed the 18-month template as misregistered: its white surface fitted its T1 image best after a 2.3-mm shift (0.5 mm in the other two templates).

2.3. Sensor arrays and helmets

The Neuromag system (306 channels) combined the sample recording's VectorView sensor geometry with the T3 coil definitions of MNE-Python (types 3014 and 3024). Its white noise was the typical value of the TRIUX datasheet, 3.5 fT/ $\sqrt{\text{Hz}}$ for the magnetometers and 3.6 fT/(cm $\sqrt{\text{Hz}}$) for the gradiometers, with an 18-mm Dewar spacing between coils and helmet surface (Elekta Neuromag, n.d.). No signal-space separation was applied.

Each OPM was a single-axis sensor whose sensing center lay nominally 7 mm above the MRI scalp (Alem et al., 2023; Osborne et al., 2018), with a few sites moved outward for clearance. Its field was averaged over a 10-mm cubic volume, a modeling choice that changes the peak field by at most 1.9 %. Its sensitive axis was the head-surface normal averaged within 15 mm of the site; with nearly radial axes, the dense array's ratio with the room field projected out (Section 2.4) would be 1.07 instead of 1.12 (Section S3).

OPMs covered the scalp above the plane through the preauricular points and a point 30 mm above the nasion, keeping at least 20 mm from either preauricular point. The dense OPM array filled this region at a minimum spacing of 17 mm (208 sites on the adult, more sites than any clinical OPM array). The site-matched OPM array placed one OPM at each Neuromag site whose projection onto the scalp fell inside it (98 of 102 sites; Fig. 1A–C). On every smaller head, the dense array was rebuilt by the same rules (144–174 sites), and the site-matched array took that head's Neuromag sites at top contact (80–91 sites; Table 1).

Clinical systems reach about 100–128 channels with dual- or triaxial sensors at fewer sites (Feys et al., 2025; Shen et al., 2026). We did not model such arrays; a triaxial array at the site-matched sites (294 channels) served as a control for the gain from extra axes. The OPM white noise was 15 fT/√Hz in the primary analysis, flat over 1–40 Hz, and was varied over 7, 10, 15, 20 and 30 fT/√Hz (the clinical studies state their noise figures without a band).

The fixed adult helmet kept the adult’s sensors and noise for every head (Fig. 1D). In its primary placement, top contact, the head was taken from the pose that puts its head frame where the adult’s was recorded and raised until the nearest magnetometer coil was 20 mm from the scalp, a best-fit assumption that favors Neuromag (Gaetz et al., 2008; Tesan et al., 2010). A family of 12 source-blind placements, chosen from the geometry alone (contacts, shifts and rotations; Section S5), gave a placement band.

To separate head size from helmet fit, we also placed each head in a helmet fitted at the adult’s gap. The head was moved sideways onto the helmet’s midline (laterally centered), and the helmet was scaled about the head origin until the median magnetometer-to-scalp gap equaled the adult’s in the same pose, 29.6 mm. For comparison, the adult’s gap is 29.8 mm at its measured head position and 28.4 mm at top contact. No coil could come within the 18-mm Dewar spacing of the scalp; in child C, this limit bound first, at 32.8 mm. None of these constructions models a pediatric SQUID system.

We compared the two helmets by the within-head contrast, the median over a head’s targets of the target-wise difference in D between the fixed and the fitted helmet, and by the interaction, a smaller head’s contrast minus the adult’s. Both compare two Neuromag helmets around the same OPM array, so the OPM terms cancel target by target and neither depends on the assumed OPM noise. Three secondary controls were a variant fitted at the adult’s top-contact gap (28.4 mm), a helmet scaled with the head, which scales the gap too, and an equal OPM site count (Section S5). For the last, the adult’s dense array was subsampled to each smaller head’s number of sites.

2.4. Noise model, validation and sensitivity analyses

Every array saw identical noise sources in one band of 1–40 Hz. Each array’s oracle covariance was computed in closed form as

$$\mathbf{C} = \mathbf{\Sigma}_n + \sigma^2 \mathbf{L} \mathbf{A} \mathbf{L}^\top + \mathbf{E} \mathbf{\Sigma}_e \mathbf{E}^\top, \quad (3)$$

where $\mathbf{\Sigma}_n$ is the diagonal covariance of white sensor noise; $\mathbf{L}$ holds the lead fields of 1,755 independent cortex-normal background dipoles on a 7-mm grid, and $\mathbf{A}$ is the diagonal matrix of their Voronoi-cell areas; and $\mathbf{E}$ and $\mathbf{\Sigma}_e$ are the array’s responses to eight external-field terms (three uniform and five linear-gradient components, the room field) and their coefficient covariance. The background is a cortically constrained form of the random-dipole model of spontaneous activity (de Munck et al., 1992): each background dipole’s moment variance is σ^2 times its area ($\sigma^2 = 0.0354 \text{ nAm}^2 \text{ per mm}^2$). Like the targets, the background omitted the cortex within 4 mm of the inner skull.

Two quantities were fitted to the sample dataset: $\mathbf{\Sigma}_e$, to its empty-room recording, and the scalar σ^2 , so that the median good gradiometer’s modeled brain noise matched the measured one (a root-mean-square value of 37.1 fT/cm). Apart from the declared sensor noise of both systems, everything else is a prediction. We analyzed four noise conditions: sensor noise only; sensor plus brain noise, the primary condition; sensor, brain and room noise; and the last with the room field’s subspace removed from every array by one noise-weighted projection. This last condition, the room field projected out, was named as the secondary condition before any simulation.

To validate the model, we compared its Neuromag covariance with the measured covariance of the same recording and recomputed Neuromag’s detectability with the measured one (Sec-

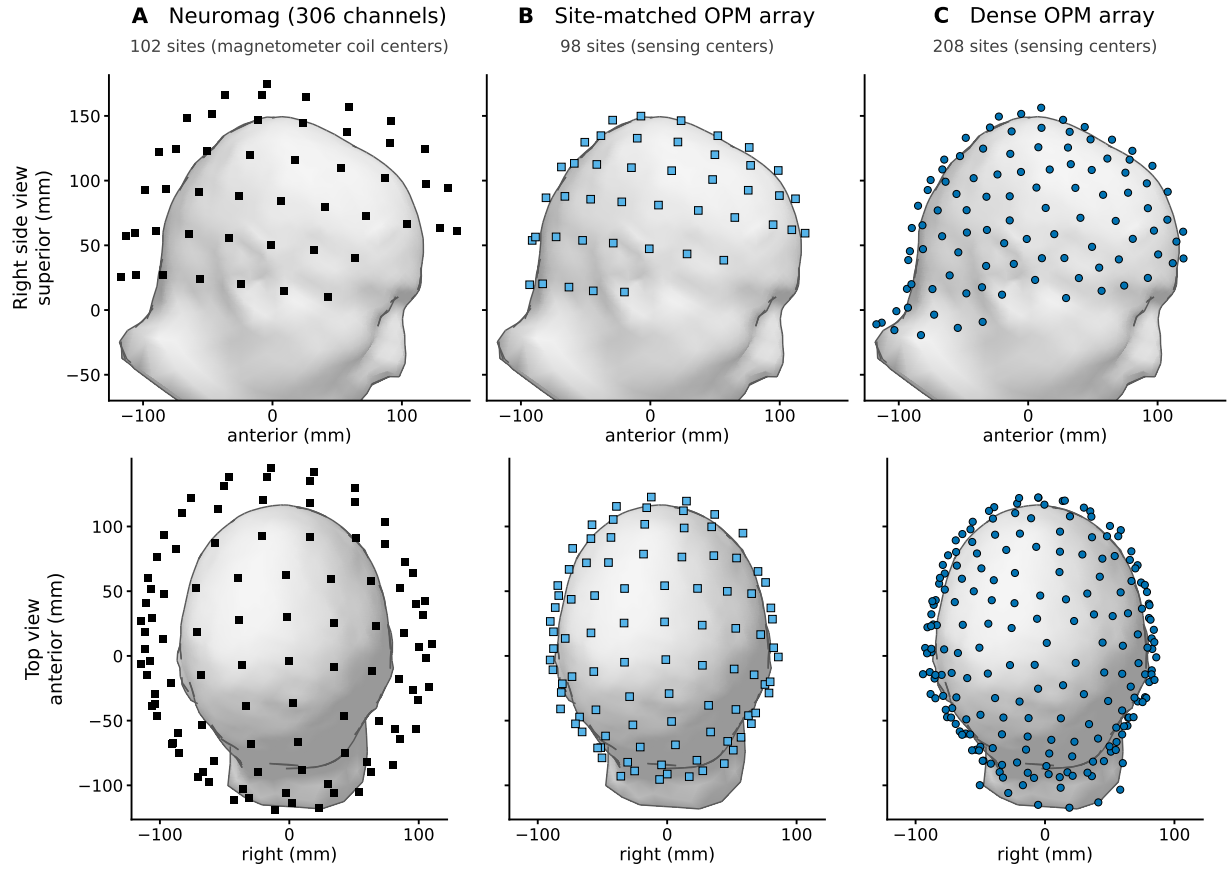


Fig. 1. Sensor arrays and helmets. (A–C) The arrays on the adult head (MNE sample subject), viewed from the right (sensors on the right half of the head) and from above. (A) Neuromag (306 channels): 102 sites, each holding one magnetometer and two planar gradiometers, at the adult’s measured head position (median 29.8 mm from the scalp). (B) Site-matched OPM array: the 98 Neuromag sites inside the OPM coverage region, projected onto the scalp. (C) Dense OPM array: 208 sites at least 17 mm apart. OPM sensing centers lie nominally 7 mm from the scalp and measure the field normal to the head surface.

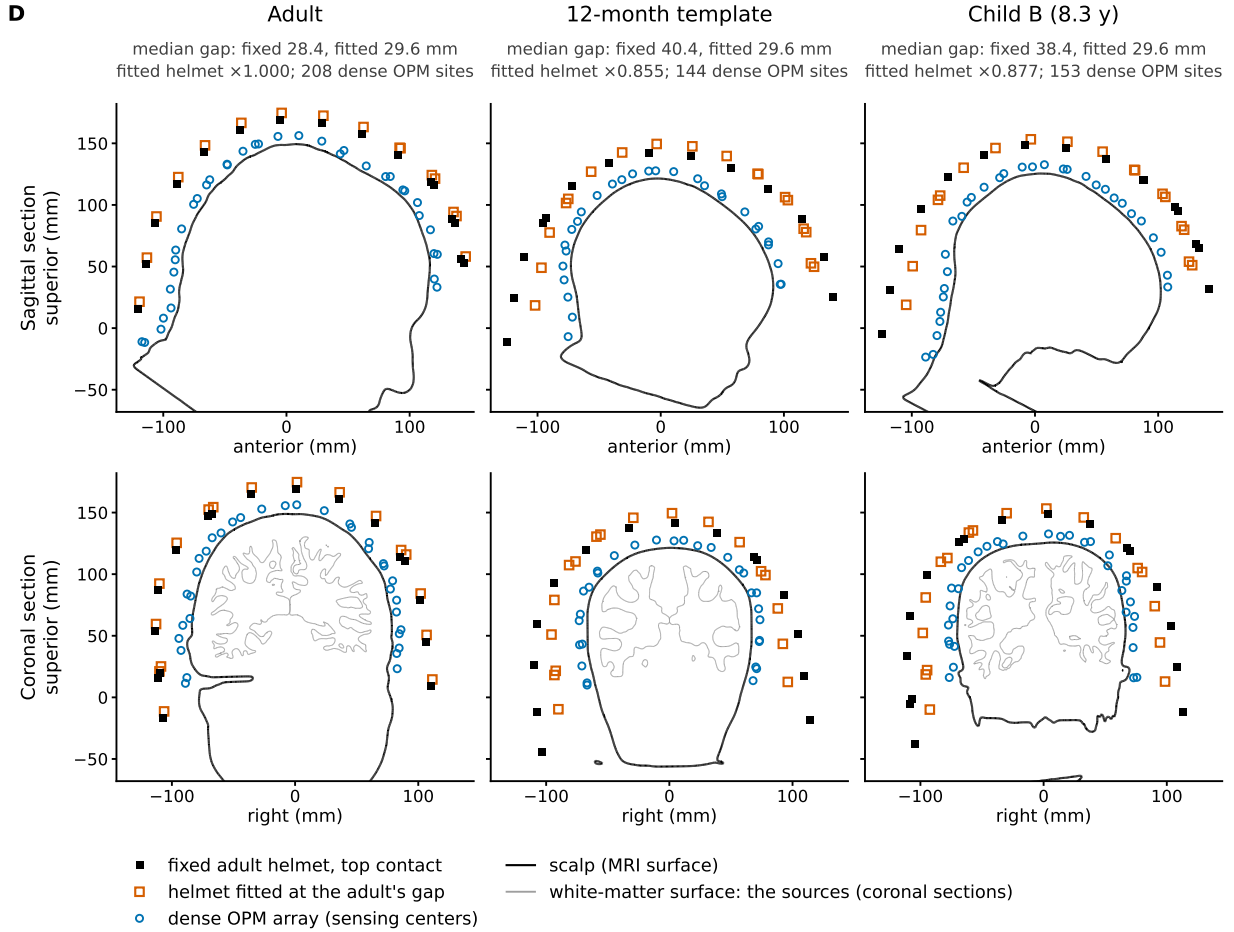


Fig. 1. (Continued.) (D) Sagittal and coronal sections through the head origin of the adult, the 12-month template and child B, at one scale: scalp (black) and, in the coronal sections, white-matter surface (gray); filled black squares, the Neuromag magnetometers of the fixed adult helmet at top contact; open vermilion squares, the helmet fitted at the adult's gap (scale factor above each column); open blue circles, the dense OPM sites; only sensors within 20 mm of the section are drawn. Median magnetometer-to-scalp gap: 28.4, 40.4 and 38.4 mm in the fixed helmet; 29.6 mm in the fitted helmet. The children's modeled skull is shown against their MRIs in Fig. S6A. Infant template: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016); child B: OpenNeuro ds005234 (Fadeev et al., 2024, 2025).

tion S3). Because the OPM’s covariance cannot be measured here, we computed the implied OPM/Neuromag ratio under five stated assumptions about how each system’s real noise departs from the model (scenarios A–E; Section 3.2). Each sensitivity analysis changed one element of the adult model and recalibrated σ^2 . The cortex 2–4 mm from the inner skull was added to the background, to the targets, or both. The OPM noise was colored ($1/f$ plus white noise, with corners at 1, 3 and 10 Hz) under a matched filter that also whitens across frequency. Idealized cardiac and ocular far-field sources were added, and all of these changes were combined.

2.5. Simulated interictal spikes

The spike study ran the noise model in the time domain, in segments shared by all arrays, with the spike-wave complex of Hunold et al. (2016) stretched to three morphologies as the spike. Two matched-filter detectors read the data, both whitened with a Ledoit–Wolf covariance (Ledoit and Wolf, 2004) estimated from 10 min of null data rather than with the oracle covariance. The oracle detector knew the source and the time and tested at a per-trial false-positive probability of 0.001. The practical detector knew neither: it scanned candidate field patterns on a 10-mm grid that did not contain the true sources, with thresholds for a nominal 1 false event per minute frozen on independent null data. Both detectors are idealized, because they use the true forward model and the simulated waveform family. S50, the dipole strength for 50 % detection, was interpolated linearly in log strength between the tested strengths.

In the exploratory run, spikes were placed at 72 locations per anatomy, stratified by orientation and by depth in four bands from 10–20 to 45–70 mm (18 locations each). Each location received 18 focal events of 10–320 nAm and patch events (cortical patches of 10-mm radius), each in its own noise realization and identical for every array. In this and the confirmatory run, the adult sat at its measured head position, where its D is 0.28 dB higher than at top contact (Section 3.4), and the smaller heads at top contact; every comparison was made within one anatomy.

The confirmatory design was declared in a configuration file committed to the public code repository the day before the run, which used the same code and configuration; only the repository’s history attests this, and the design was not registered externally. The endpoint compared the dense OPM array with Neuromag (306 channels) under the practical detector at a nominal 1 false event per minute, for focal spikes 10–20 mm below the scalp in all nine anatomies. The threshold, set on 20 min of calibration null data, rests on about 20 false events, a Poisson uncertainty of about 22 %.

The test was a two-sided sign-flip test on the per-location differences in detection counts, Holm-adjusted over the anatomies (Holm, 1979) at $\alpha = 0.05$, with the paired S50 ratio r as the effect size, so that the OPM’s 50 % detection strength is lower by $1 - 1/r$. The declared test estimated p by Monte Carlo when more than 20 location differences were nonzero; we report exact p values over all sign patterns, which give the same decisions in all 25 families of Holm-adjusted tests (the endpoint and the secondary analyses; Section S7).

The confirmatory run drew 36 new locations per anatomy and five noise replicates per event, the first carrying the endpoint, and used separate null data for threshold fitting and for rate evaluation. The 36 locations doubled the exploratory 18 per band after two pilot runs in child A on seeds not used for the analysis (locations favoring the OPM and Neuromag, 8 vs 1 and 9 vs 4; $p = 0.074$ and 0.21). The pilots were not designed to estimate power. Secondary analyses were a declared mismatched detector (off-template morphologies, candidate patterns from a single-compartment BEM and a shared coregistration error of 2 mm and 2°), the site-matched array, the oracle detector and rate-matched thresholds re-fitted on held-out null data.

In the exploratory run, we localized detected spikes at 24 locations by equivalent-current-dipole fits and by dynamic statistical parametric mapping (dSPM; Dale et al., 2000) at the true

peak sample, through a single-compartment BEM with a coregistration error of 2 mm and 2° (Section S7).

2.6. Statistical analysis

Neighboring targets are correlated, so intervals are 95 % percentile bootstrap intervals over the adult's 70 atlas labels or over each smaller head's parcels, with 1,000 resamples for the primary comparisons and 200 for the adult's sensitivity analyses (every count in Table S16). They describe spatial variation within one anatomy, not variation between subjects or head placements. We call a ratio established when its interval excludes 1, and a difference in dB when its interval excludes 0. In the spike study, the location was the unit of resampling and testing. Table 2 summarizes the estimands; apart from the within-head contrast and the interaction, all of them depend on the assumed OPM noise. The exploratory spike endpoint (chosen from 24 comparisons per OPM array), the primary helmet placement and the regions of interest were named after the whole-cortex results were known, and the model was revised while results were known (Section S8). Only the confirmatory run carries a pre-specified test.

2.7. Software

Simulations used MNE-Python 1.13.2 (Gramfort et al., 2013) and NumPy 2.5.3 (Harris et al., 2020); the random seeds are listed in Table S17. The analysis code was written and run with the assistance of a generative AI tool (Claude, Anthropic), as declared at the end of the article.

Table 2. Estimands.

Estimand	Definition	Unit; interval	Section
Adult ratio	Median over all 7,661 targets (medial wall included) of the paired ratio $d_{\text{OPM}}/d_{\text{Neuromag}}$; D is $20 \log_{10}$ of it	ratio; parcel bootstrap	3.1–3.3
Scenario ratio	The adult ratio with sensor, brain and room noise (Neuromag’s 305 channels, one bad channel excluded) under one of five assumptions about how each system’s real noise departs from the model, informed by Neuromag’s measured covariance (scenarios A–E)	ratio; parcel bootstrap	3.2
Break-even noise	OPM white noise at which the adult ratio equals 1	fT/ $\sqrt{\text{Hz}}$; parcel bootstrap	3.1, 3.2
D of a head	Area-weighted median of D over cortical targets, medial wall excluded	dB; parcel bootstrap within the head	3.4, 3.5
Δ	$D_{\text{head}} - D_{\text{adult}}$, the adult placed by the same helmet rule; vertex-wise for the scaled adults, the adult-area-weighted median of parcel differences for templates and children	dB; parcel bootstrap	3.4, 3.5
Within-head contrast ^a	Area-weighted median over one head’s targets of the target-wise difference $D(\text{fixed helmet}) - D(\text{fitted helmet})$, not the difference of the two medians	dB; parcel bootstrap	3.5
Interaction ^a	A smaller head’s within-head contrast minus the adult’s (the adult placed by the same rule in both helmets), taken per vertex or parcel as for Δ and summarized by its area-weighted median; not $\Delta(\text{fixed}) - \Delta(\text{fitted})$ of the medians	dB; parcel bootstrap	3.5
Residual at top contact	Δ of a head in the helmet fitted at the adult’s top-contact gap, against the adult at top contact in the fixed helmet	dB; parcel bootstrap	3.5
Δ at equal site count	Δ with the adult’s dense array subsampled to the smaller head’s number of OPM sites, both in the fitted helmet	dB; parcel bootstrap	3.5
Placement band	Δ at each feasible one of 12 source-blind placements of the fixed helmet, as a difference of medians; crossed band: any feasible placement of the smaller head against any of the adult’s	dB; range	3.4
Peak-channel SNR	Each system’s largest ratio of a channel’s noise-free signal to its own noise standard deviation; OPM over Neuromag	ratio or dB; parcel bootstrap (adult) or median only (smaller heads)	3.1, 4.1, 4.3
Mean-power SNR	Channel mean of squared SNRs (Goldenholz et al., 2009), compared as an amplitude ratio	ratio; parcel bootstrap	3.1, 4.3
S50 ratio	Dipole strength for 50 % detection, Neuromag over OPM, from the location-pooled detection curves	ratio; location bootstrap	3.6

d : known-topography detectability with the oracle covariance, Eq. (1); D : Eq. (2). Neuromag: all 306 channels unless stated. Top contact: the primary placement of the fixed adult helmet; fitted helmet: the helmet fitted at the adult’s gap (Section 2.3). Intervals: 95 % percentile bootstrap intervals over atlas parcels (or over simulated spike locations), describing spatial variation within one anatomy. At the adult’s measured head position, the dense array’s ratio is 1.14 ($D = +1.17$ dB) under the adult convention and 1.16 (+1.27 dB) under the area-weighted convention of the smaller heads. ^aA property of Neuromag alone: independent of the assumed OPM noise.

3. Results

3.1. Determinants of the adult comparison

With sensor noise alone, Neuromag led both OPM arrays: the median paired ratio over the 7,661 targets was 0.74 for the dense array and 0.53 for the site-matched array (Fig. 2). Brain noise changed the comparison. With the room field included, brain noise made up 92 % of a dense-array OPM channel's in-band variance and 74 % of a Neuromag magnetometer's, whose room-field share was 25 %. With sensor plus brain noise, the dense array's ratio was 1.14 [1.12, 1.17] ($D = +1.17$ dB), and the OPM was ahead at 7,657 of the 7,661 targets. The site-matched array's ratio was 1.01 [0.99, 1.03] ($D = +0.08$ dB), with no established advantage.

Which Neuromag channels and which SNR measure entered the comparison changed the number. Against the 204 gradiometers alone, the ratios were 1.33 (dense) and 1.13 (site-matched), and against the 102 magnetometers 1.19 and 1.04. By peak-channel SNR, the closest analogue of the sensor-level SNRs of clinical studies (Section S1), the dense array was ahead of the magnetometers (1.23 [1.17, 1.27]) but behind the best of all 306 channels (0.90 [0.88, 0.93]). By mean-power SNR, the dense array's advantage was small (1.04) and the site-matched array's was 1.05 [1.04, 1.06].

The OPM noise floor could remove or reverse the advantage. From 7 to 30 fT/√Hz, the dense array's ratio fell from 1.30 to 1.01 [0.99, 1.03], and the site-matched array's from 1.08 to 0.93. The break-even white noise was 31.5 [28.5, 34.9] fT/√Hz for the dense array and 16.7 [14.0, 19.4] fT/√Hz for the site-matched array (Fig. 3D).

Over the ranges tested, other single changes moved the dense ratio less (Fig. 2). Projecting out the room field gave 1.12 [1.08, 1.15], and the site-matched array then trailed Neuromag (0.95 [0.90, 0.98]). A 3- or 6-mm gap between OPM and scalp lowered the dense ratio to 1.09 or 1.04, and 30 fT/√Hz with a 3-mm gap reversed it (0.97 [0.96, 0.98]). A triaxial array at the site-matched sites recovered much of the dense advantage (1.12; 1.05 with tangential axes twice as noisy).

With a covariance estimated from 4,680 independent samples, equivalent in time and bandwidth to 60 s of data, the ratios were 1.16 [1.14, 1.18] (dense) and 1.03 [1.01, 1.05] (site-matched), and a 3–70 Hz band changed little (1.14 and 1.01). Across these conditions, then, the OPM noise floor or the array could reverse which system led, the comparator and the metric decided which reported number showed it, and depth changed the size of the dense advantage and the sign of the site-matched array's (Section 3.3).

3.2. The noise model against measured Neuromag noise

The model predicted 0.73 of the magnetometers' measured brain-noise amplitude, an independent prediction (Fig. S2), or 0.89 with the heart's field removed. The gradiometer level at randomly held-out sites (1.00; [0.93, 1.09] over 1,000 splits) and the overall pattern of channel correlations agreed with the measurement, the former nearly by construction. The modeled brain noise was, however, too flat from inferior to superior sites (1.34 and 0.75 when fitted on the inferior or the superior half), and its magnetometer correlations and per-channel variances were poorly predicted. The magnetometers' excess over the model lay in a few far-field components (Section S3).

For detectability, this excess favored Neuromag: with its measured covariance, Neuromag's detectability was 1.14 [1.08, 1.20] times its modeled value (Fig. 3A), a factor as large as the dense array's whole advantage. What this implies for the OPM depends on its real noise, which no measurement here constrains (Fig. 3B).

If the model's error was common to both systems (scenario A), the ratios remained 1.14 and 1.01. They remained close to these values when both cortical backgrounds were scaled by the

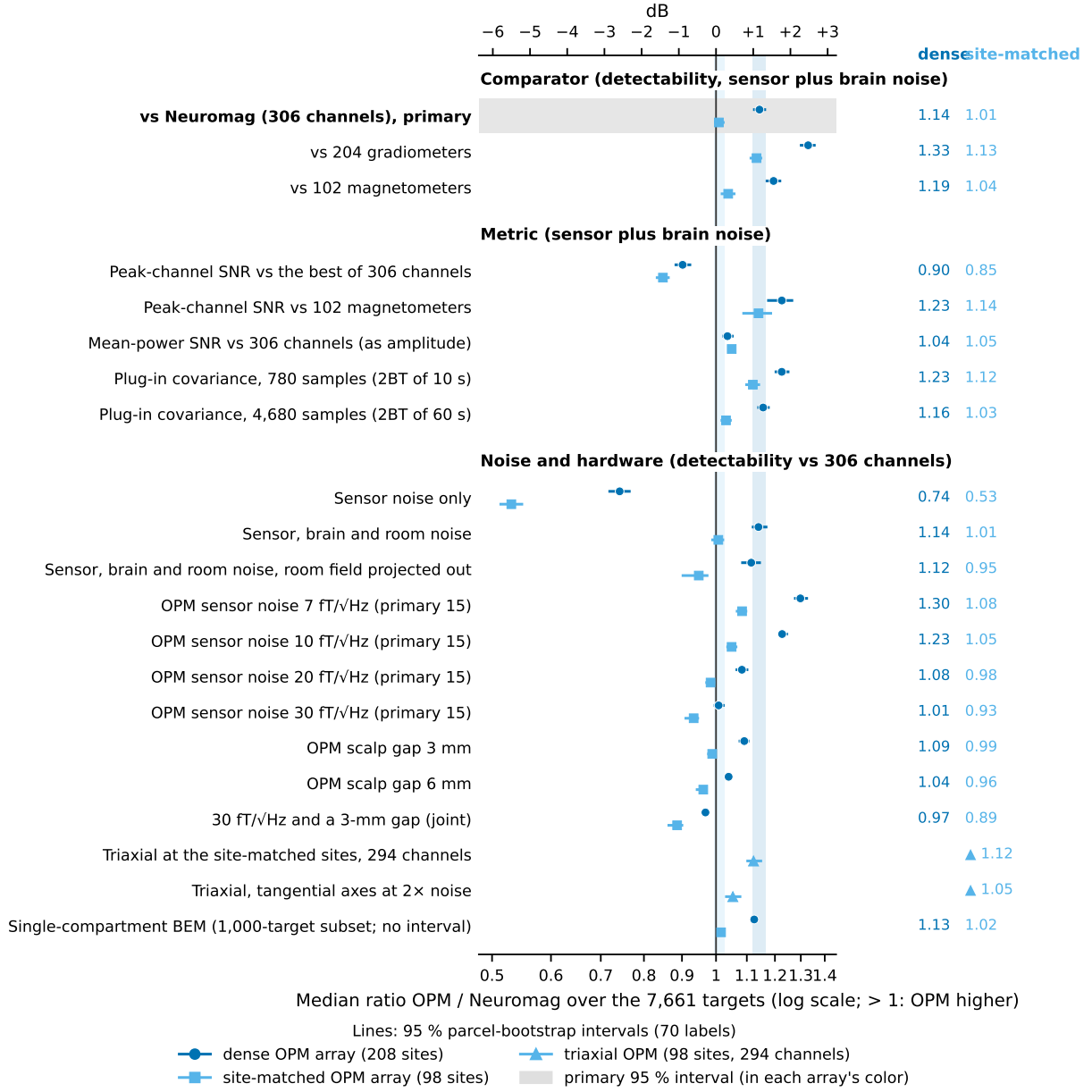


Fig. 2. Determinants of the adult comparison. Median over the 7,661 targets of the paired detectability ratio OPM/Neuromag (306 channels) for the dense OPM array (circles), the site-matched OPM array (squares) and a triaxial array at the site-matched sites (triangles, 294 channels), with 95 % parcel-bootstrap intervals over 70 atlas labels (resample counts in Table S16); logarithmic axis, with D (dB) on the top axis. The gray row and the bands shaded in each array's color mark the primary condition: oracle covariance, sensor plus brain noise, OPM noise of 15 fT/√Hz, no scalp gap, three-layer BEM and the adult at its measured head position. Each other row changes one factor, except the joint row (30 fT/√Hz with a 3-mm gap). Comparator: Neuromag's 306 channels, gradiometers or magnetometers. Metric: peak-channel SNR; mean-power SNR as an amplitude ratio; and plug-in covariances from 780 or 4,680 independent samples (2BT, twice the bandwidth B times the duration T , for $T = 10$ and 60 s). The single-compartment BEM row uses a subset of targets and has no interval. The columns on the right give the medians.

magnetometers' variance excess over the model, a factor of 1.86 (scenario B), or when Neuromag's measured empty-room noise was combined with the modeled brain noise (scenario E). With Neuromag taken as measured and the OPM as modeled (scenario D), the dense ratio was 1.03 [0.99, 1.08], with no established advantage, and the site-matched ratio 0.89. With the OPM bearing the whole magnetometer variance excess as cortical noise (scenario C, the most pessimistic), the ratios were 0.80 and 0.68.

The omissions we could test moved the advantage little at 15 fT/√Hz (Fig. 3C; Table S4). The cortex 2–4 mm from the inner skull gave 1.14 when added as background and 1.15 when added as targets. A $1/f$ component with a 10-Hz corner gave 1.07 under the primary detector and 1.08 under one that also whitens across frequency, and cardiac and ocular sources changed the dense ratio by -1.6% to $+1.6\%$. The joint run of near-skull background and the 10-Hz corner, with frequency whitening (Fig. 3D, dashed), gave 1.08 [1.06, 1.11] at 15, 1.016 [0.997, 1.035] at 20 and 0.93 at 30 fT/√Hz. Under the joint run, the advantage was thus established up to 15 but not at 20 fT/√Hz, against up to 20 (1.08 [1.06, 1.11]) but not 30 fT/√Hz under the primary model.

3.3. Depth and brain regions

The dense advantage was largest near the scalp (Fig. 4A): the ratio was 1.60 [1.55, 1.65] in the 10–15 mm bin, 1.13 [1.12, 1.14] at 25–30 mm and 1.05–1.07 from the 35–40 to the 55–60 mm bin, every interval above 1. The site-matched array had an equal-SNR depth even with sensor plus brain noise: its intervals lay above 1 down to the 25–30 mm bin and below 1 from 40–45 mm. With the room field projected out, the dense array's bin median fell below 1 from 45–50 mm and the site-matched array's from 25–30 mm, with intervals below 1 only at 55–60 mm and from 30–35 mm, respectively.

For the dense array, the fall with depth was geometric. It was present with sensor noise alone (Fig. 4A, dotted) and in the peak field itself (Fig. 4B), and brain noise multiplied the ratio by a nearly constant factor in every bin (1.51–1.68). On the cortex (Fig. 4C), the dense array was ahead almost everywhere, most of all on the lateral convexity.

By region (Fig. 5), the dense advantage was largest in the frontal lobe (1.22 [1.18, 1.26]) and smallest in the cingulate (1.05 [1.04, 1.07]) and the insula (1.09). Of the regions chosen from the epilepsy literature, the precentral gyrus favored the dense array most (1.21) and the parahippocampal cortex least (1.12), although 12 atlas parcels, mostly cingulate and medial, had smaller advantages. The site-matched array trailed most on ventral and polar surfaces, such as the frontal pole (0.71), the medial and lateral orbitofrontal cortex (0.78 and 0.79) and the temporal pole (0.83), and, among the literature's regions, in the mesial temporal cortex (parahippocampal and entorhinal; 0.92).

These rankings are exploratory point estimates under the modeled noise. Under scenario D (Neuromag as measured, the OPM as modeled), the dense ratio became 0.75 in the cingulate, 0.96 in the parietal, 1.01 in the frontal and 1.32 in the temporal lobe (point estimates). Under the same scenario, which includes the room field, the dense bin medians fell below 1 from the 25–30 mm bin.

3.4. Smaller heads in the fixed adult helmet

The dense array's D was $+1.19$ to $+2.13$ dB in the smaller heads and $+1.00$ [$+0.82$, $+1.17$] dB in the adult at top contact (Fig. 6A, C). At top contact, the median magnetometer-to-scalp gap was 32.6–40.4 mm in the smaller heads and 28.4 mm in the adult (Table S5). Top contact brings Neuromag closer to the adult's head than its measured position does, where D is $+1.27$ dB (area-weighted; Table 2). Every smaller head's point estimate of Δ was positive (Fig. 6B), at $+0.44$ to $+1.07$ dB in the scaled adults and infant templates. It was smaller in the school-aged children,

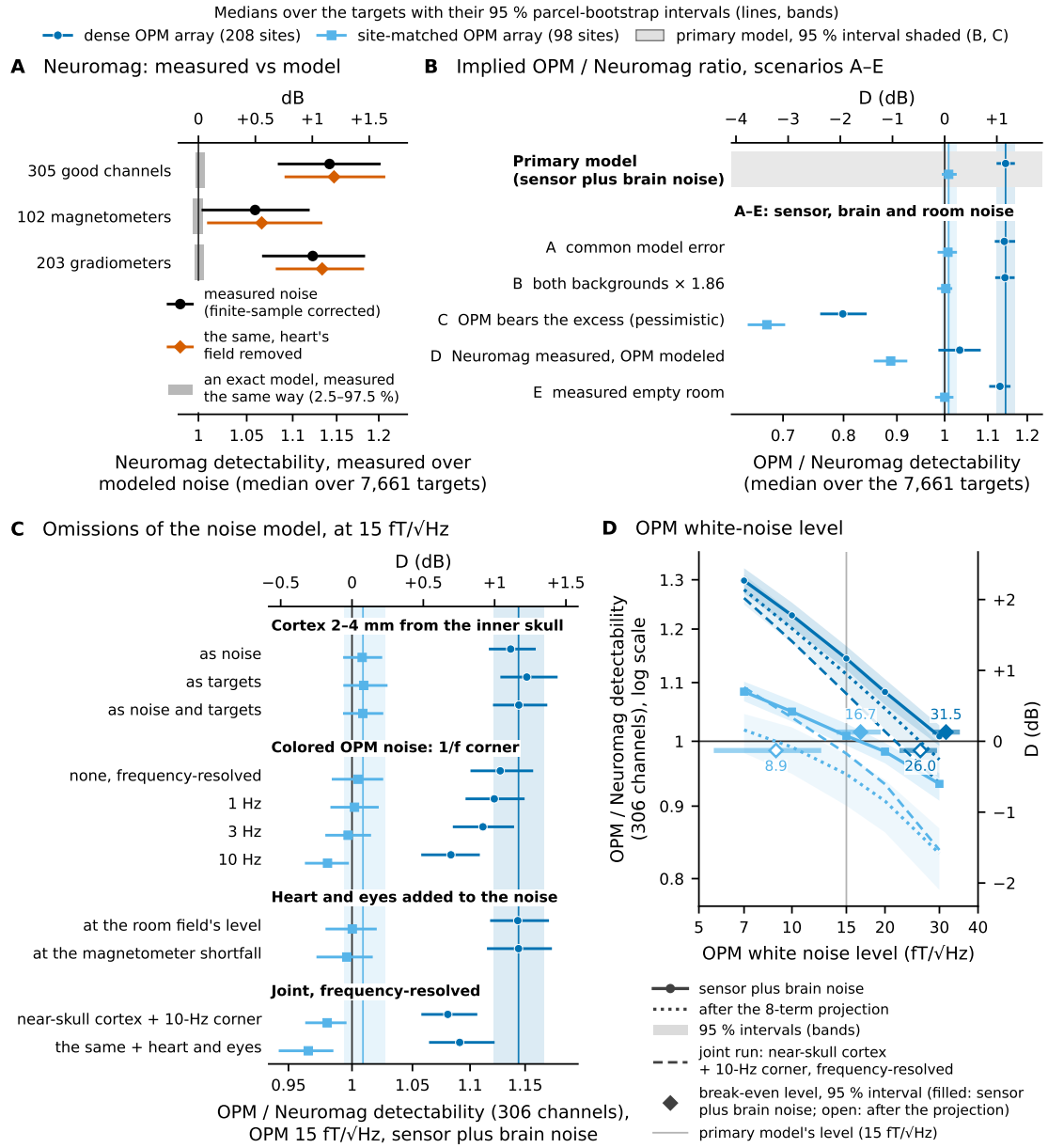


Fig. 3. The noise model against measured Neuromag noise. Adult at its measured head position; medians over the 7,661 targets with 95 % parcel-bootstrap intervals (lines, and bands in each array's color); dense OPM array, circles; site-matched OPM array, squares. Ratio axes are logarithmic, with dB on the top or right-hand axis. (A) Neuromag's detectability with its measured noise covariance over that with the modeled one (sensor, brain and room noise; finite-sample corrected) for its 305 channels (one bad channel excluded), the magnetometers and the gradiometers (black), the same with the heart's field removed (orange), and the range an exact model would give when measured in the same way (gray). (B) The implied OPM/Neuromag ratio under scenarios A–E of Section 3.2; gray row, the primary model. (C) The ratio at 15 fT/√Hz when what the model leaves out is added: cortex 2–4 mm from the inner skull as noise, targets or both; colored OPM noise (1/f corners at 1, 3 and 10 Hz); cardiac and ocular sources; and the joint runs. (D) The ratio against the OPM white-noise level with sensor plus brain noise (solid), the room field projected out (dotted) and the joint run of C (dashed), with the break-even levels (diamonds with intervals; filled, sensor plus brain noise; open, projected) and the primary level (vertical line).

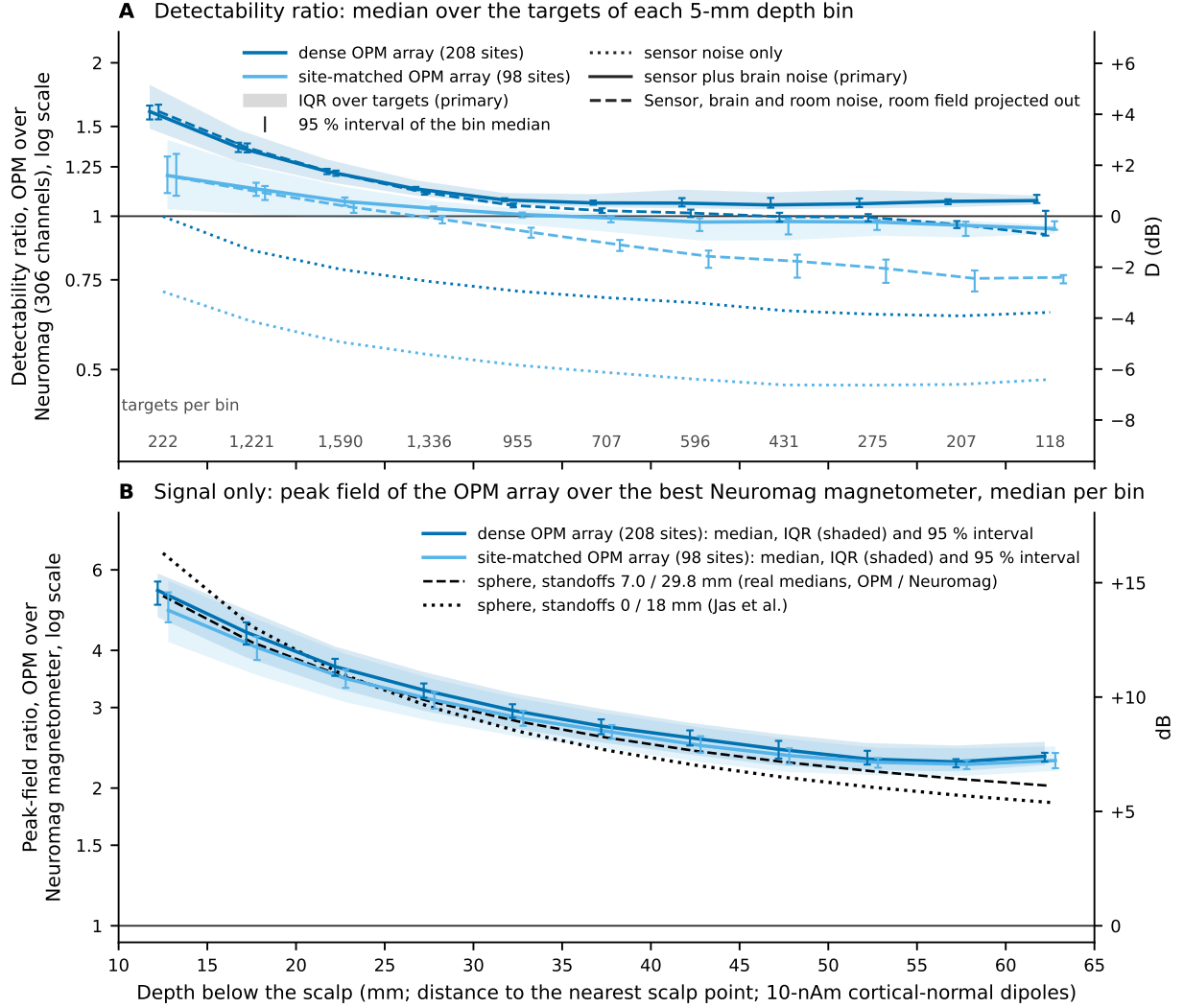


Fig. 4. The adult comparison by depth and on the cortex. (A) Median paired detectability ratio OPM/Neuromag (306 channels) in 5-mm depth bins for the dense (208 sites) and site-matched (98 sites) OPM arrays, with sensor noise only (dotted), sensor plus brain noise (solid; the primary condition, its interquartile range, IQR, over targets shaded) and the room field projected out (dashed). Bars: 95 % intervals of the bin median from 1,000 resamples of the parcel labels in the bin. Gray numbers: targets per bin (the few targets deeper than the last bin are not drawn). (B) Signal only: peak field of the OPM array over that of the best Neuromag magnetometer (median, IQR and 95 % interval per bin), with the analytical sphere at the real median standoffs (dashed) and at the standoffs of Jas et al. (2026) (dotted). Ratio axes are logarithmic, with dB on the right.

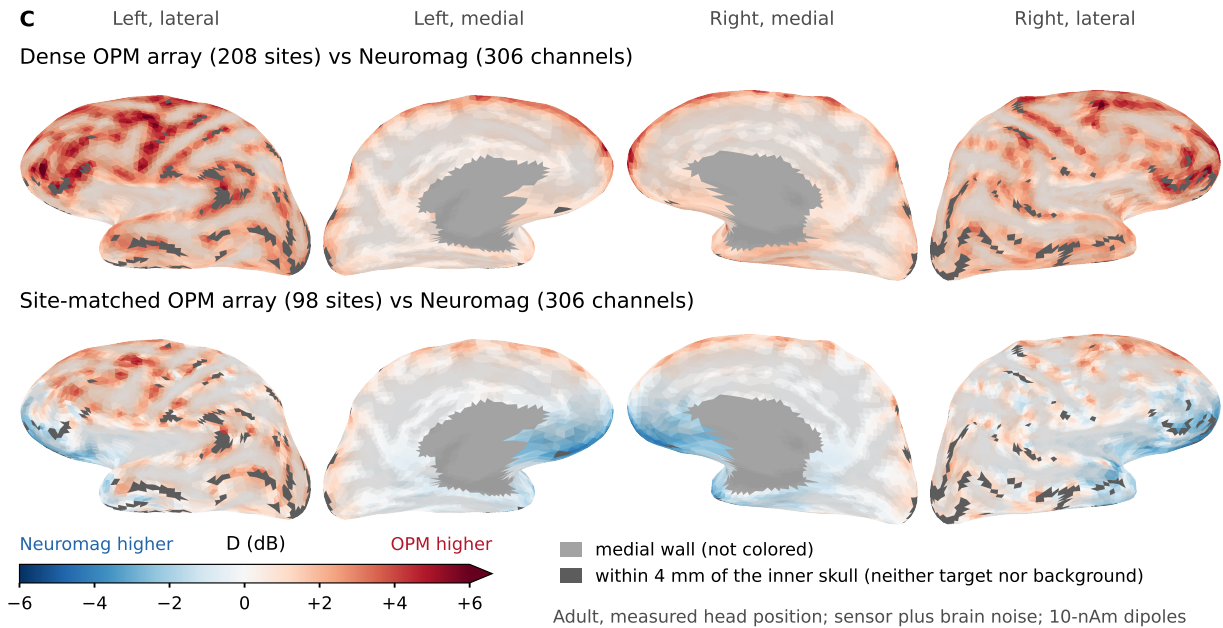


Fig. 4. (Continued.) (C) D on the adult's inflated cortex at its measured head position, sensor plus brain noise, for the dense (top) and site-matched (bottom) arrays against Neuromag's 306 channels; red, the OPM's detectability is higher. Median D is +1.17 dB (dense) and +0.08 dB (site-matched) over all 7,661 targets. Light gray, medial wall (its 558 targets enter the medians but are not colored); dark gray, cortex within 4 mm of the inner skull, neither a target nor part of the background.

provisional examples (Section 2.2): +0.30 [+0.11, +0.38], +0.41 [+0.28, +0.54] and +0.17 [−0.07, +0.34] dB in A, B and C, with child C's interval including zero.

Placement, which the parcel intervals exclude, was the larger uncertainty. Over the source-blind placements of the fixed helmet, Δ spanned +0.20 to +1.12 dB in the school-age scaled adult and +0.06 to +0.37, +0.19 to +0.63 and −0.10 to +0.70 dB in children A, B and C. It was positive at every placement in every head except child C, although the crossed bands included zero in every child (Section S5).

3.5. Helmet fit versus head size

We answer this question with one primary quantity, the interaction: how much more the fixed helmet favors a smaller head than the adult, measured against the helmet fitted at the adult's gap (Fig. 7). Because it compares two Neuromag helmets around the same OPM array, it does not depend on the assumed OPM noise. The interaction was +0.40 to +1.10 dB in the scaled adults and templates, every interval above zero (per head in Table S6), and +0.40 to +1.11 dB with the helmet fitted at the adult's top-contact gap. In each of these heads, the fitted helmet restored to Neuromag what the fixed helmet's gap and placement took away: the within-head contrast (Fig. 7D) was +0.17 to +0.76 dB at the adult's gap and +0.27 to +0.91 dB at its top-contact gap.

The interaction exceeds the heads' own within-head contrasts because the adult's contrast is negative, −0.23 dB: the fitted helmet holds the adult laterally centered, a looser pose than top contact. At the adult's top-contact gap, its contrast is −0.11 dB, the effect of the construction itself.

Of the excess advantage that the scaled adults and templates showed in the fixed helmet (+0.44 to +1.07 dB), little remained. Fitted at the adult's gap, their Δ against the adult fitted the same way was +0.02 to +0.15 dB (point estimates; interval bounds −0.09 to +0.33 dB). Every template interval included zero, and only the school-age scaled adult's lay just above it; the highest value

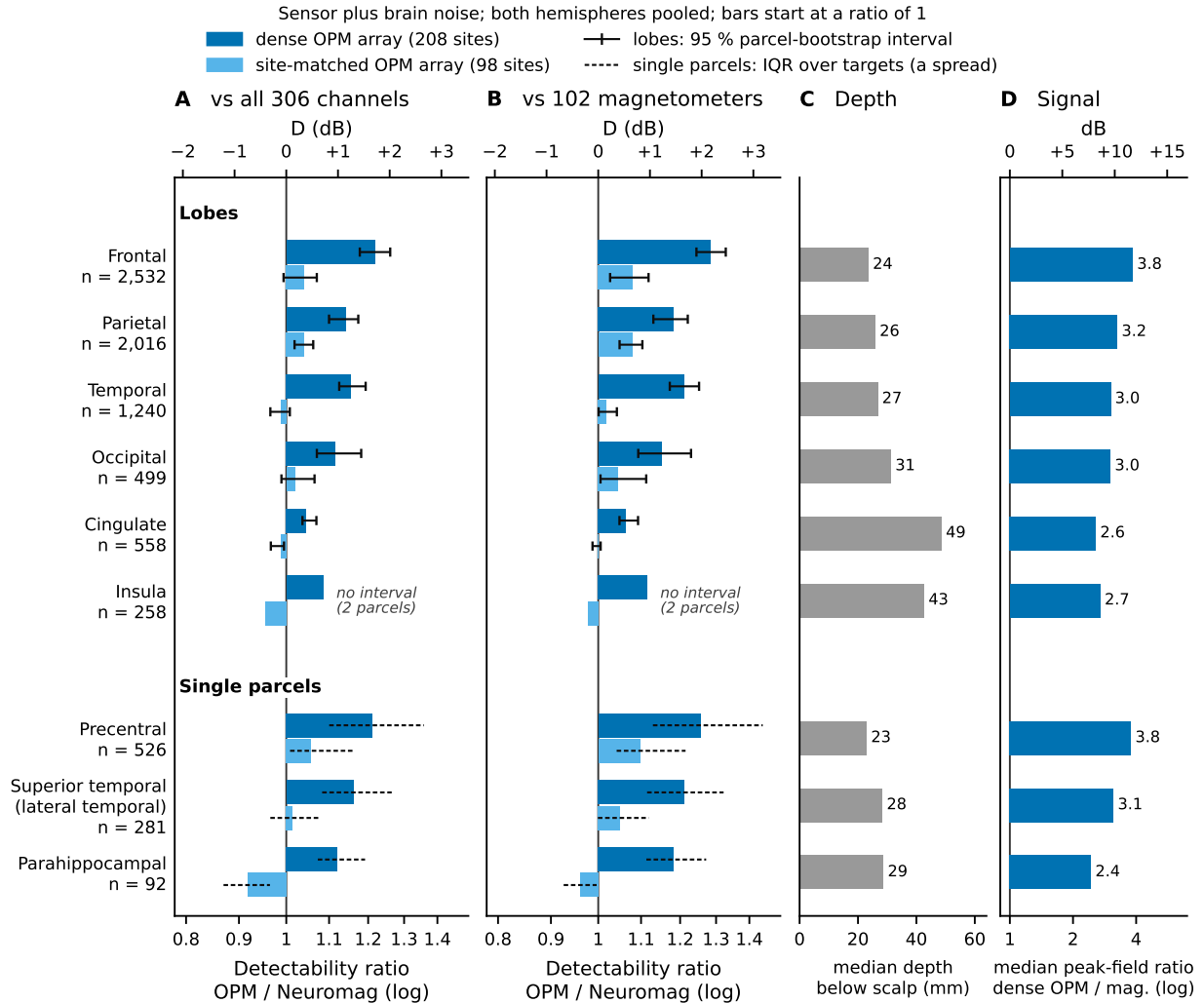


Fig. 5. The adult comparison by brain region. Adult at its measured head position; sensor plus brain noise, OPM noise of 15 fT/√Hz; hemispheres pooled. (A) Median detectability ratio OPM/Neuromag (306 channels) for the dense and site-matched arrays; bars start at 1 on a logarithmic axis, with D (dB) on the top axis. Lobes: median with 95 % parcel-bootstrap intervals (solid; the insula, with two parcels, has none). Single parcels (precentral; superior temporal, labeled lateral temporal; parahippocampal): median with the interquartile range over the parcel's targets (dashed), a spread rather than a confidence interval. (B) The same against the 102 magnetometers alone. (C) Median depth below the scalp. (D) Median peak-field ratio of the dense array over the best magnetometer (signal only). Targets per region: 2,532 frontal, 2,016 parietal, 1,240 temporal, 499 occipital, 558 cingulate, 258 insula, 526 precentral, 281 superior temporal and 92 parahippocampal.

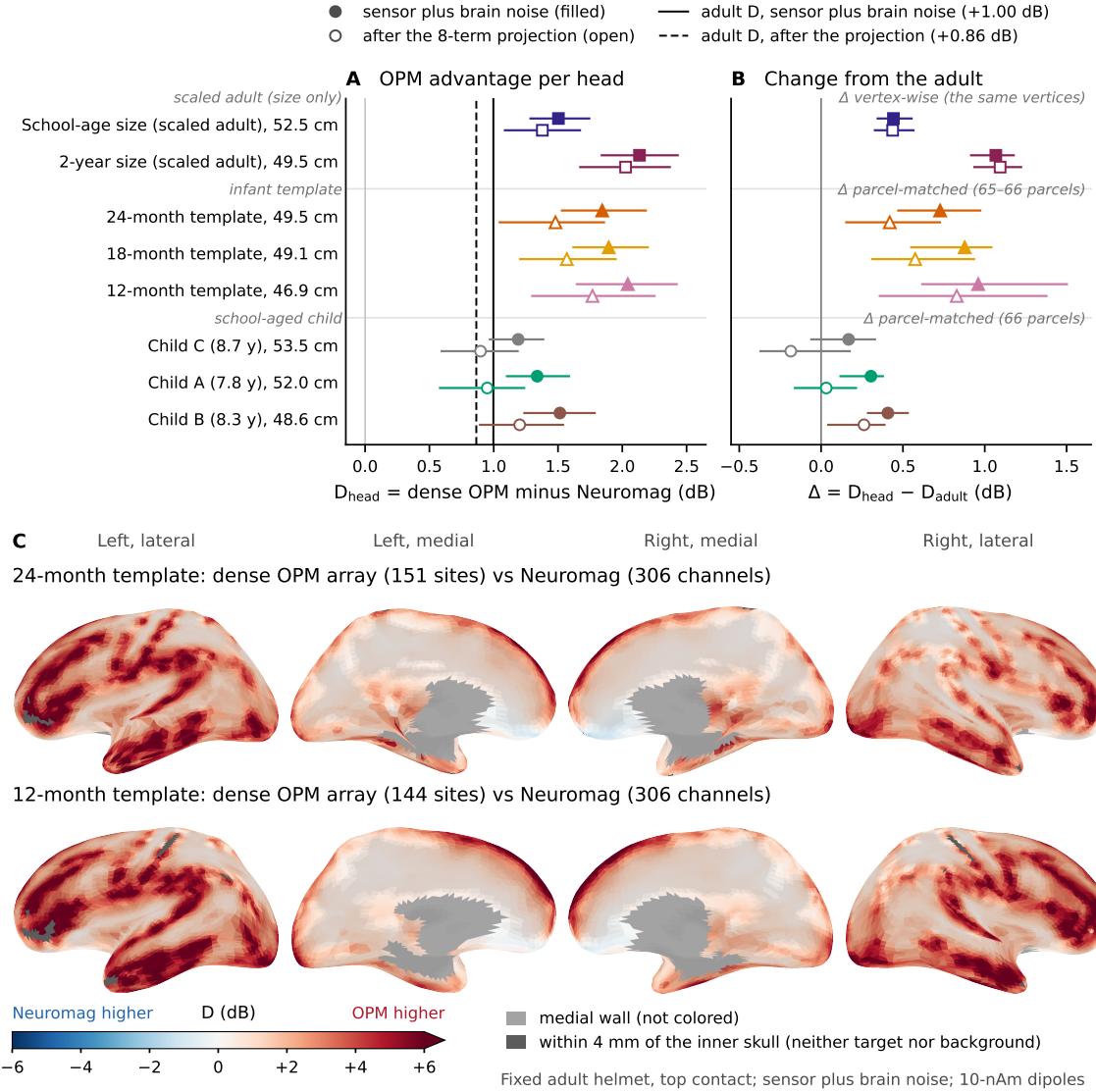


Fig. 6. Smaller heads in the fixed adult helmet. Dense OPM array against Neuromag (306 channels) at top contact; D is the area-weighted median over cortical targets, medial wall excluded. (A) D for the eight smaller heads with sensor plus brain noise (filled) and with the room field projected out (open), with 95 % parcel-bootstrap intervals; vertical lines, the adult's D at top contact (+1.00 and +0.86 dB). (B) $\Delta = D_{\text{head}} - D_{\text{adult}}$ with 95 % parcel-bootstrap intervals (1,000 resamples; spatial variation within one anatomy, without between-subject or placement variability), vertex-wise for the scaled adults and over parcels for templates and children. Heads are grouped by class and ordered by head circumference (the number after each label); squares, scaled adults; triangles, infant templates; circles, children (provisional examples); one color per head, as in Fig. 8. (C) D on the inflated cortex of the 24-month and 12-month infant templates (dense arrays of 151 and 144 sites, top contact, sensor plus brain noise), on the color scale of Fig. 4C; values beyond ± 6 dB take the end color (3.3 % and 6.1 % of the cortical targets lie above +6 dB). Area-weighted median D : +1.84 and +2.04 dB. The 18-month template (in A and B) is classed misregistered by the MRI check (Section 2.2). Infant templates: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016); children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025).

was that of the 18-month template, which the MRI check classed as misregistered. Fitted at the adult's top-contact gap, these heads kept +0.10 to +0.23 dB against the adult at top contact (4 of 5 intervals above zero). With the room field projected out, every smaller head's point estimate at the adult's gap lay below the adult's (−0.67 to −0.05 dB; three of the eight intervals including zero).

In the children, provisional examples (Section 2.2), Δ at the adult's gap was negative in A and B (−0.30 [−0.48, −0.22] and −0.39 [−0.54, −0.17] dB). Fitted at the adult's top-contact gap and set against the adult at top contact, however, it was not established (−0.11 [−0.40, +0.06] and −0.22 [−0.35, +0.01] dB; Section S5), and their within-head contrasts were +0.29 and +0.42 dB. In child C, whose two gaps nearly coincide (Fig. 7A), Δ was +0.01 [−0.23, +0.18] dB, and its within-head contrast, −0.12 [−0.20, −0.02] dB, matched the effect of the construction itself (−0.11 [−0.15, −0.08] dB in the adult at an equal median gap).

None of these quantities apportion the effects exactly, because the fitted helmet changes the gap distribution, the placement and the sensors' coverage together. In the fitted helmet and at equal OPM site count, Δ was +0.16 to +0.49 dB (Section S5). In the scaled adults and templates, their fewer OPM sites offset a similar amount, whereas in children A and B no such gain was established (+0.16 [−0.12, +0.23] and +0.16 [−0.10, +0.41] dB). In this model, fitting the helmet to the adult's median gap thus removed most of the excess advantage that the scaled adults and templates showed in the fixed adult helmet.

3.6. Simulated interictal spikes

The declared confirmatory test was significant in all nine anatomies (Fig. 8A; Table S11). The S50 ratio Neuromag/OPM was 1.27–1.47, every interval above 1, and every Holm-adjusted p over the 36 newly drawn locations per anatomy was at most 0.010; the dense array thus detected these spikes at 21–32 % lower strength. The nine anatomies are not independent replications (Section 2.2). The ratios were of the order of the adult's detectability ratio at these depths (1.36–1.60; Fig. 4A) and below it by point estimate in the 24-month template, child A, child B and child C. In the children, the OPM's advantage at 10–30 mm was below the adult's by point estimate in 11 of 12 depth strata (Section S5).

The test was significant in all nine anatomies in each of the five noise replicates (ratio standard deviation, 2.9–6.4 %; pooled ratios, 1.22–1.44), and with the oracle detector, which gave 1.27–1.62 (Fig. 8B). On separate null data that no threshold had seen, the frozen thresholds gave 0.90–2.15 false events per minute for the dense array and 0.70–2.25 for Neuromag. An approximate equal-rate test (uncorrected; the arrays share their background and room noise) gave $p < 0.05$ in two anatomies, the 24-month template and child B (Section S7). With each threshold re-fitted on held-out null data to the nominal 1 false event per minute (realized rates 0.90–1.60 and 0.70–1.45; $p < 0.05$ in one anatomy, the scaled adult of school-age size), the ratios were 1.27–1.50. The test was then significant in all nine anatomies.

The mismatched detector changed the ratio by a factor of 0.91–1.11, every interval including 1, and its test was significant in all nine anatomies with frozen thresholds (Fig. 8C) and in 8 of 9 with rate-matched ones (child B, Holm-adjusted $p = 0.072$). The site-matched array's lower S50 was significant in 5 of 9 anatomies, none of them a child (Table S11); its ratios were 0.99–1.30, above 1 by point estimate in 8 of 9, in line with its shallow advantage (Section 3.3).

The exploratory run (18 of its 72 locations per anatomy in this band; endpoint chosen after the analyses) gave concordant results (S50 ratios 1.18–1.65); its deeper bands and discordant events are reported in Section S7. Both runs used the fixed helmet and the primary noise model, so they do not test whether the gap drives a detection benefit in smaller heads. Spikes were not simulated under scenario D, under which the dense array's detectability ratio at these depths stays at 1.20–1.35 in the adult (Section S3).

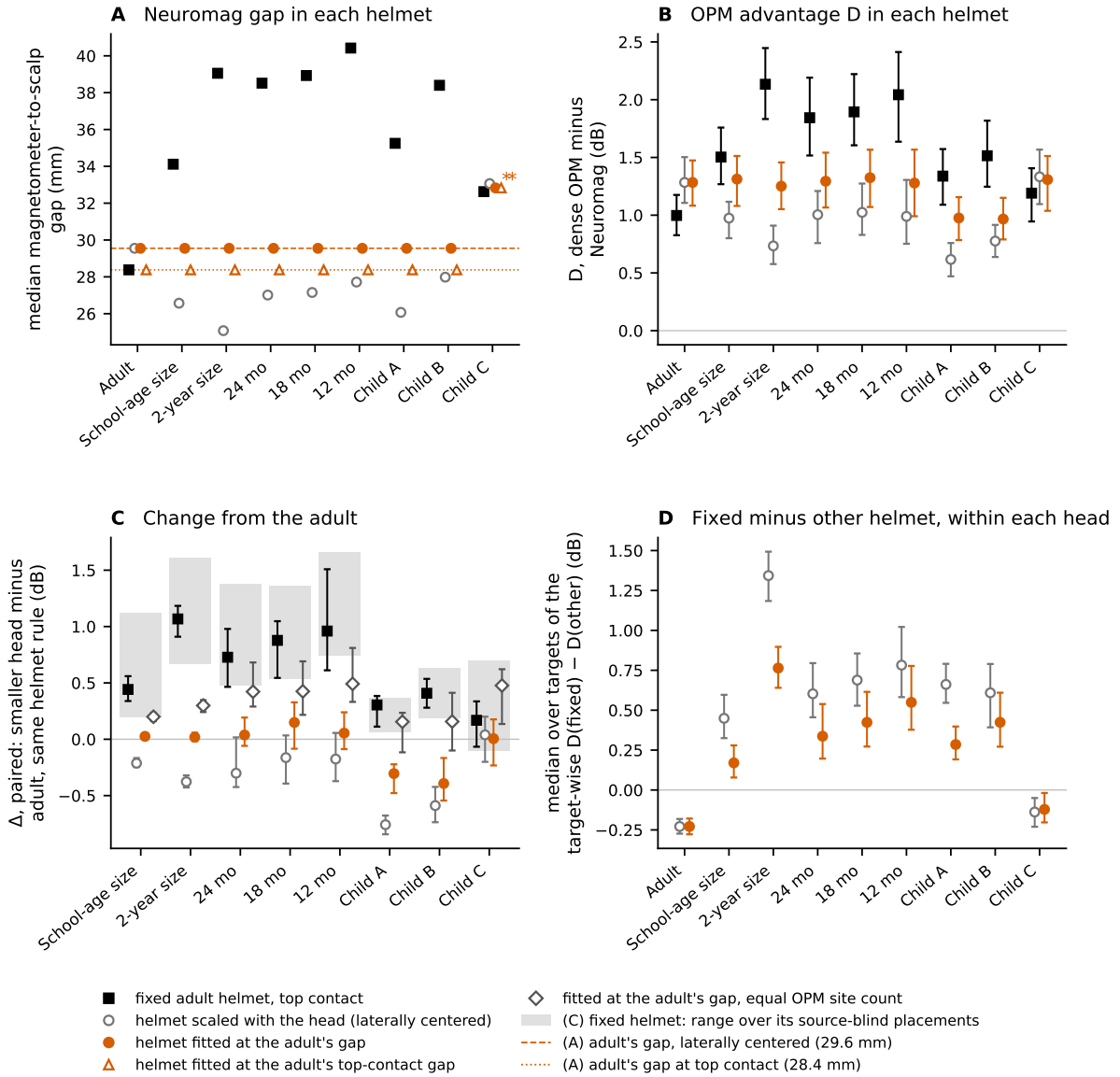


Fig. 7. Helmet fit versus head size. Dense OPM array against Neuromag (306 channels), sensor plus brain noise; D is the area-weighted median over cortical targets; bars, 95 % intervals from parcels resampled within each head. Helmets: the fixed adult helmet at top contact (filled black squares); the helmet fitted at the adult's gap of 29.6 mm (filled vermilion circles; Section 2.3; asterisks, child C, where the 18-mm limit binds); its variant fitted at the adult's top-contact gap of 28.4 mm (open vermilion triangles, in A only); and the helmet scaled with the head (open gray circles). Each head keeps its own OPM arrays. (A) Median gap in each helmet; dashed and dotted lines, the adult's gap when laterally centered and at top contact. (B) D in each helmet. (C) Δ against the adult placed by the same rule; open dark-gray diamonds, the fitted helmet at equal OPM site count; gray ranges, Δ at each feasible one of the 12 source-blind placements of the fixed helmet (11 in child C). (D) The within-head contrast between the fixed helmet and the fitted helmet (filled) or the helmet scaled with the head (open). Head labels: School-age size and 2-year size, the scaled adults; 24, 18 and 12 mo, the infant templates; the children are provisional examples and the 18-month template is classed misregistered (Section 2.2).

394 No localization difference was established in the exploratory run. Dipole fits of detected 320-
395 nAm focal spikes had median errors of 3.7–7.6 mm in every array and anatomy, of the order of the
396 coregistration displacement (2.1–3.2 mm). Of 360 paired comparisons of dipole and dSPM errors
397 and of joint detection and localization, 65 reached $p < 0.05$ uncorrected (about 18 expected by
398 chance; 61 favoring an OPM array). Of these, 8 survived a Bonferroni correction within anatomy
399 and array, and 0 survived one over all 360. At 320 nAm, the paired dipole-fit differences lay
400 within a few millimeters of zero, and the dSPM differences were negative (OPM closer) or zero,
401 depending on the implementation (Section S7; Fig. S14).

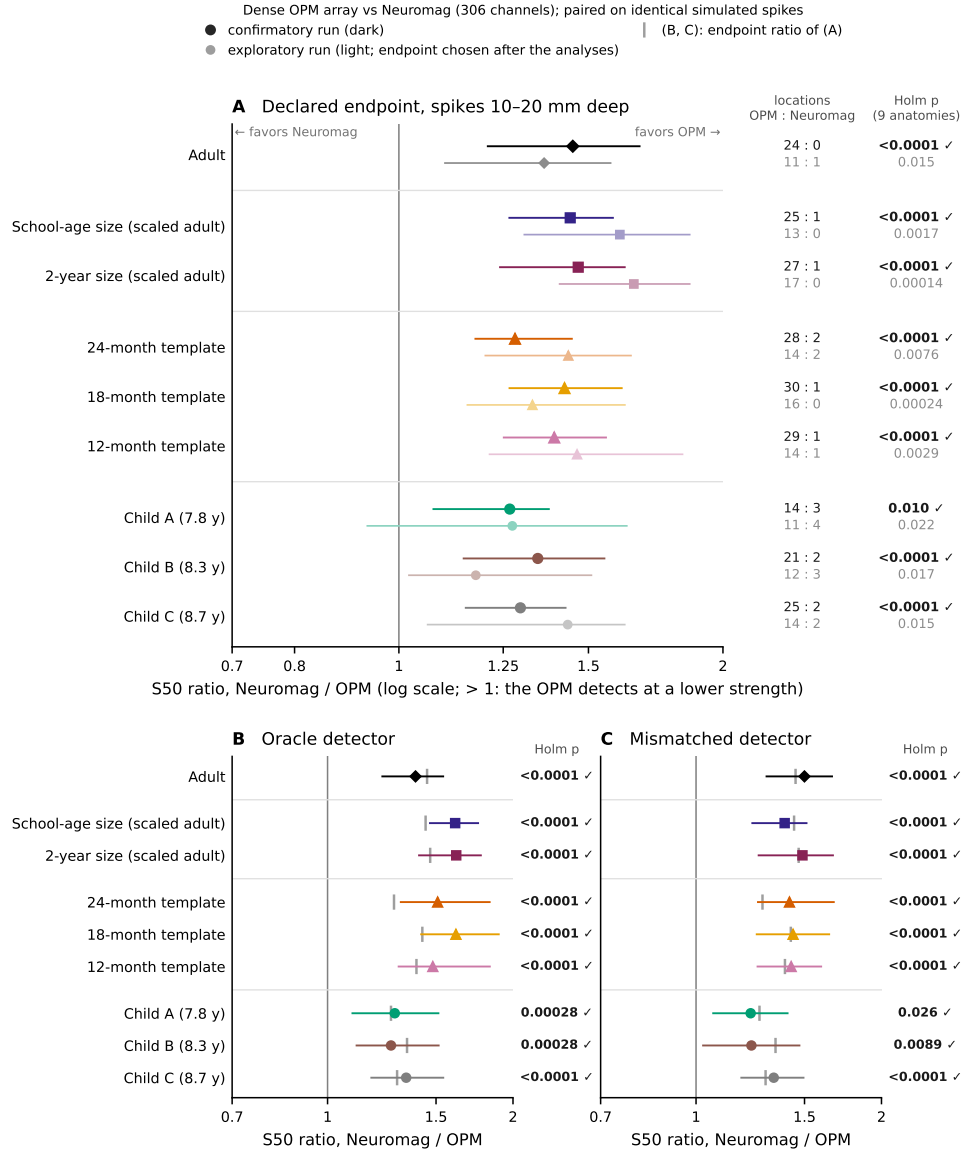


Fig. 8. Simulated interictal spikes: the confirmatory run. S50 ratio (S50, the dipole strength detected in 50 % of events), Neuromag (306 channels) over the dense OPM array, for focal spikes 10–20 mm below the scalp at 36 newly drawn locations per anatomy (648 focal events; the first of five noise replicates); logarithmic axis (above 1, the OPM detects at a lower strength); lines, 95 % location-bootstrap intervals (1,000 resamples). (A) The declared endpoint: the practical scanning detector, idealized in that it uses the true forward model and the simulated waveform family, with thresholds frozen for a nominal 1 false event per minute (dark markers). Right: the numbers of locations at which the OPM or Neuromag detected more events (the others are ties) and the two-sided sign-flip p, exact and Holm-adjusted over the nine anatomies (bold with a check mark, below 0.05). Light markers: the exploratory run (18 locations per anatomy; endpoint chosen after the analyses). (B) The oracle detector (known source, waveform and time; per-trial false-positive probability 0.001). (C) The mismatched detector (Section 2.5). In B and C, secondary analyses, gray bars mark the endpoint's ratio of A. Diamond, the adult at its measured head position; other heads at top contact in the fixed adult helmet, with symbols and colors as in Fig. 6; the children's rows are provisional examples.

4. Discussion

In a realistic adult head with explicit modeled noise, a dense array of 15-fT/ $\sqrt{\text{Hz}}$ OPMs had a modestly higher detectability than the Neuromag system, whereas an array at Neuromag’s own sites showed no established advantage. The advantage depended on the OPM noise floor and on how the comparison was made, and it was as large as the model’s error for Neuromag’s noise. In smaller heads, most of the extra advantage that the OPM gained in the fixed adult helmet disappeared when the helmet was fitted to each head at the adult’s gap. With an idealized detector, the modeled noise and the fixed adult helmet, the dense array detected weaker superficial spikes than Neuromag in all nine anatomies (not independent replications), without an established localization difference (exploratory).

4.1. Noise floor, array design and the noise model

The spherical model of Jas et al. (2026) places the OPM advantage within an equal-SNR depth set by an assumed noise ratio. With explicit noise, the dense array has an advantage at every depth and shows an equal-SNR depth only after the room projection (by bin medians, with an interval below 1 only at 55–60 mm) or with Neuromag’s measured noise (scenario D, point estimates). The site-matched array keeps an equal-SNR depth even with sensor plus brain noise. Depth therefore changes the size of the dense advantage and the sign of the site-matched array’s. The OPM noise floor or the array, in contrast, can reverse which system leads: the site-matched array falls behind Neuromag at 30 fT/ $\sqrt{\text{Hz}}$, and the dense array’s advantage is no longer established there and reverses with a 3-mm gap.

The fall of the dense advantage with depth is geometric. It appears without brain noise and in the peak field itself, and brain noise scales the ratio by a nearly constant factor at every depth. Brain noise turns the SQUID advantage into an OPM advantage because it raises the noise of a Neuromag magnetometer about 9.3-fold over its low sensor floor but that of an OPM only about 5.7-fold (Section S3). The OPM’s higher sensor noise therefore matters less once brain noise dominates. The comparator and the metric then decide how a reported number shows the difference: a peak-channel SNR against magnetometers favors the OPM, one against the best of all channels favors Neuromag, and a channel-averaged SNR shows little difference.

How far the dense advantage can be trusted depends on the OPM’s real noise. The model predicted 0.73 of Neuromag’s measured magnetometer brain-noise amplitude without being fitted to the magnetometers, but its error for Neuromag’s detectability was as large as the dense advantage. Depending on how the OPM shares that error, the five scenarios placed the dense ratio between 0.80 and 1.14. The advantage survived the omissions we could test, although together they lowered the noise level up to which it was established from 20 to 15 fT/ $\sqrt{\text{Hz}}$. The dense advantage is therefore a prediction rather than a measurement, and its break-even OPM white noise, 31.5 [28.5, 34.9] fT/ $\sqrt{\text{Hz}}$ (site-matched array, 16.7 [14.0, 19.4] fT/ $\sqrt{\text{Hz}}$), is what a measured OPM noise covariance in the same band would test directly.

The regional pattern is exploratory and rests on the modeled noise. Under that noise, the dense advantage is largest on the frontal and central convexity and smallest in cingulate cortex, and the site-matched array trails on ventral and polar surfaces, including the mesial temporal cortex. With Neuromag’s measured noise (scenario D, point estimates), the frontal advantage falls to about 1, the cingulate and parietal lobes fall below 1, and the temporal lobe leads.

4.2. Head size and helmet fit

The sphere predicts a larger OPM benefit in smaller heads because a shell at a constant standoff leaves more of a small brain within the equal-SNR depth (Jas et al., 2026). In the fixed adult helmet, every smaller head’s point estimate of Δ was indeed positive. Fitting the helmet to each

head at the adult's gap, however, removed most of this excess in the scaled adults and templates (an interaction of +0.40 to +1.10 dB, every interval above zero). The interaction compares two Neuromag helmets around the same OPM array, so, unlike every other estimand of this study, it does not depend on the unmeasured OPM noise. In this model, then, the sphere's prediction holds in the adult helmet but largely reflects how that helmet fits a smaller head, its gap, placement and coverage together. Because the fitted helmet changes all three, it does not isolate the median gap, or exclude head size, as a cause.

What remained in the fitted helmet depends on the adult's reference placement and, for the templates, on their scalp convention. Against the adult fitted the same way, it was +0.02 to +0.03 dB in the scaled adults and +0.04 to +0.15 dB in the templates (the upper end that of the misregistered 18-month template), and against the adult at top contact +0.10 to +0.23 dB. The conclusion does not rest on the templates. The two scaled adults keep the adult's anatomy, and their fixed-helmet excess of +0.44 and +1.07 dB fell to +0.13 and +0.11 dB against the adult at top contact when the helmet was fitted at the adult's top-contact gap. They also gave the lowest and the highest interactions of the scaled adults and templates (+0.40 and +1.10 dB; templates, +0.54 to +0.70 dB).

At equal OPM site count, Δ was +0.16 to +0.49 dB (not established in children A and B), the quantity comparable to the sensor-noise-only result of Zahran et al. (2022); in the scaled adults and templates, their fewer OPM sites offset a similar amount. For pediatric MEG, these results suggest that much of the extra on-scalp advantage that small heads show in an adult helmet reflects that helmet's fit. Our fitted helmet keeps the adult's sensors and does not model a pediatric SQUID system, such as one built for young children (Tesan et al., 2010), and how a child is positioned in an adult helmet matters as well (Gaetz et al., 2008). The children's values are provisional examples: two of them fell below the adult fitted the same way, a difference not established in the top-contact comparison.

4.3. Simulated spikes and the clinical literature

In the confirmatory run, the dense array detected superficial simulated spikes at 21–32 % lower strength than Neuromag in all nine anatomies, and the ranking survived the declared detector mismatch. The S50 ratios were of the order of the adult's detectability ratio at these depths, so the detectability difference carried over to detection with an idealized detector. The run kept the fixed adult helmet and the primary noise model, so it does not show whether helmet fit drives a detection benefit in smaller heads, nor whether the benefit would survive the OPM noise of scenario D. No localization difference was established; dipole-fit errors were of the order of the coregistration error, which may have dominated the comparison.

The clinical reports and the model are compatible, but the model cannot identify which factor operated in any study (Table 3). Higher or similar OPM SNR against SQUID magnetometers (Feys et al., 2022; mixed results in Feys et al., 2025) corresponds to the model's peak-channel SNR against the magnetometers, which favors the OPM in every head (Section S6). Equal or lower SNR against planar gradiometers, or against the nearest channels of few or noisier sensors (Hillebrand et al., 2023; Schwartz et al., 2025), corresponds to peak-channel SNR against the gradiometers or the best channel. The larger gains reported in school-aged children in an adult helmet (Feys et al., 2022) are compatible with the fixed helmet's fit, its gap and placement together (Sections 3.4 and 3.5).

The weaker OPM results for temporal and mesial temporal sources (Feys et al., 2024, 2025; Shen et al., 2026) have two analogues in the model: deep sources are weaker for every system, and the site-matched array trails on ventral and polar surfaces. The second analogue is fragile, because the model overstates Neuromag's detectability in the temporal lobe (measured over modeled, 0.89;

Section S3). With a channel-averaged metric, the site-matched array had a small but established advantage (1.05 [1.04, 1.06]; Section 3.1) where Matsubara et al. (2026) found no region with a higher OPM SNR, with brain noise only and in patients' anatomies. A cohort's report of no significant difference in detection (Ren et al., 2025, abstract only) cannot adjudicate the model's predictions, whose endpoints differ (whole-cortex detectability and the detection of simulated superficial spikes).

By whole-cortex detectability against Neuromag's 306 channels with the oracle covariance, a single-axis array like the site-matched one (98 sites, about the channel count of the larger clinical arrays) showed no established advantage at 15 fT/ $\sqrt{\text{Hz}}$ and a SQUID advantage when noisier. Its small established advantages depended on the comparator, the metric, the covariance estimate or depth, within about 30 mm of the scalp, including simulated superficial spikes in 5 of 9 anatomies. Clinical arrays reach their channel counts with dual- or triaxial sensors at fewer sites. The triaxial control at the site-matched sites (1.12) shows what extra axes can recover there, but the model does not determine whether an array of about 64 dual-axis sites would lie above or below the site-matched array. The modest whole-cortex advantage thus belongs to an array with more sites than any used in epilepsy so far, at a white noise within the clinical range.

Table 3. Clinical OPM studies of epilepsy with a SQUID comparison or a regional result, and their qualitative model analogues.

Study: patients (ages); regions	OPM sensors	Comparator; SNR measure	Reported, OPM vs comparator	Model analogue: quantity, comparator, anatomy, value	Direction	Not matched
Feys et al. (2022): 5 children (5–11 years); extratemporal ^P	Rb, 32 single-axis, cap around the focus ^P ; noise not reported	102 SQUID magnetometers (MEGIN TRIUX ^P); peak over baseline SD of each IED at the channel where the averaged IED peaks ^P	Amplitude 2.3–4.6× in 5/5; SNR 27–60 % higher in 4/5, no difference in 1	Peak-channel SNR against the 102 magnetometers; school-age scaled adult at top contact in the adult helmet: <i>D</i> site-matched +1.66 dB, dense +2.15 dB	Same: OPM ahead	Sensor count and placement; sensor noise; real IEDs per child vs a median over all cortex; extratemporal foci; children vs the adult’s anatomy scaled to school-age size; helmet position and band not reported
Feys et al. (2023): 1 infant (5 months; corrected age 2 months); right centro-parietal	Rb, 26, cap, 4-mm gap; noise not reported	None (other children’s recordings); SNR not defined	IED amplitude 2.5 pT; SNR 11.1	No counterpart: no SQUID recording of the infant, and absolute values where the model gives ratios between systems; the model’s youngest head is the 12-month template (head circumference 46.9 cm, against 33 cm)	–	Comparator; metric; age and head size; microcephaly and hemorrhage
Hillebrand et al. (2023): 7 (3 children, 10–12 years; 4 adults); temporal, temporo-parietal, sensorimotor	Rb, 6 dual-axis (12 channels), 7–13 fT/√Hz	TRIUX neo planar gradiometers after tSSS; automatic-detector z-score per IED	0.92–0.98 ^d ; “comparable”	Peak-channel SNR against the 204 planar gradiometers, adult: site-matched 0.86 [0.84, 0.89], dense 0.92 [0.89, 0.95]	Same: Neuromag ahead	Sensor count and axes; sensor noise (7–13 vs 15 fT/√Hz); detector z-score of real IEDs; tSSS; regions vs all cortex; children and adults vs one adult; band not reported
Badier et al. (2023): 1 adult (33 years); left temporal pole (hippocampus-only spikes seen by neither)	⁴ He, 4 on the scalp, < 50 fT/√Hz on 2 axes	4 nearest magnetometers of a 248-channel 4D system; event maximum over baseline SD	Single spike: SNR 7 vs 5; amplitude 5 pT (text) or 2.5 pT (legend) vs 1.1 pT	Peak-channel SNR against the 102 magnetometers, adult, all cortex: site-matched 1.14 [1.08, 1.19], dense 1.23 [1.17, 1.27]; temporal pole, detectability against the magnetometers: site-matched 0.88, dense 1.23; hippocampus not modeled	Same by peak channel; temporal pole: same (dense), opposite (site-matched)	Sensor count; sensor noise (< 50 vs 15 fT/√Hz); nearest channels of another SQUID system vs Neuromag’s best; one spike per session; peak-channel value over all cortex; band (2–70 Hz)
Feys et al. (2024): 1 adult (20–30 years ^P); right amygdala	Rb, 48 (116 channels), rigid helmet; noise not reported	None (SEEG reference); SNR not defined	SNR 2.75 vs 11.1–16.7 in earlier neocortical OPM data (other patients)	Within one system, detectability of parahippocampal over superior temporal sources, adult: dense 0.36, Neuromag 0.36; amygdala not modeled	Same, and equally for Neuromag	Amygdala vs a cortical proxy; patients compared across studies vs regions within one head; helmet off the scalp, implants; sensor count; noise not reported

Table 3 (continued)

Study: patients (ages); regions	OPM sensors	Comparator; SNR measure	Reported, OPM vs comparator	Model analogue: quantity, comparator, anatomy, value	Direction	Not matched
Feys et al. (2025): 11 (8–63 years); temporal lobe epilepsy (anterior, medial, basal, lateral and posterior sources)	Rb, 43–53 (98–127 channels) incl. 4 face OPMs; noise not reported	102 SQUID magnetometers (system not named); peak over baseline SD after averaging similar IEDs, peak channel	SNR higher in 1, similar in 2 of 3 tested; nominally lower in the 3 with mesial or polar sources ^d ; OPM SNR temporal 9.4 vs extratemporal 14.3	Peak-channel SNR against the 102 magnetometers, adult, all cortex: site-matched 1.14 [1.08, 1.19], dense 1.23 [1.17, 1.27]; mesial temporal cortex, detectability against the magnetometers: site-matched 0.96, dense 1.18; within the dense array, temporal over frontal lobe 0.73	Mixed (study) vs OPM ahead (model, peak channel); mesial: same (site-matched), opposite (dense); within the OPM: same	Sensor count and axes, face OPMs; regional model values are detectability, not peak-channel SNR; mesial structures vs a cortical proxy; frontal lobe for extratemporal; ages 8–63 vs one adult; noise and band not reported
Ren et al. (2025) ^a : 46 (mean 23.7 years); regions not given	128-channel whole-scalp; type and noise not given	SQUID (system and channels not given); SNR not defined	Amplitude and SNR higher; IED detection not significantly different	SNR, comparator unknown: peak-channel SNR, adult, against the 102 magnetometers site-matched 1.14, dense 1.23; against the best of all 306 channels site-matched 0.85, dense 0.90. Detection: detectability against all 306 channels, site-matched 1.01 [0.99, 1.03], dense 1.14 [1.12, 1.17]	SNR: depends on the comparator; detection: not comparable	Comparator, SNR definition, sensors, noise, band and regions not reported; per-patient detection by review vs single-dipole detectability; 46 patients vs one adult
Schwartz et al. (2025): 7 + 1 adults (22–48 years; 41 years); 5 focal temporal, 2 generalized; deep mesial (SEEG case)	⁴ He, 4 on the scalp, < 45 fT/√Hz on 2 axes	4 nearest channels of a 275-channel CTF system, type not named (4D magnetometers for the SEEG case); per-IED maximum over 20-s baseline SD	Amplitude 2.3× (group); SNR 6.72 vs 8.37; deep spike 6.8 vs 9.0	Peak-channel SNR against the best of all 306 channels, adult: site-matched 0.85 [0.83, 0.87], dense 0.90 [0.88, 0.93]; detectability with 30-fT/√Hz OPMs, the top of the model’s range: site-matched 0.933, dense 1.008	Same: Neuromag ahead or no established difference	Sensor count; sensor noise above the model’s range; nearest channels of an unnamed type vs Neuromag’s best channel; per-IED SNR in sequential sessions; temporal and generalized foci vs all cortex; band (6–70 and 3–70 Hz)
Shen et al. (2026): 64 dual-mode (128 channels), 51 temporal, 17 extratemporal	64 dual-mode (128 channels), < 15 fT/√Hz, rigid helmet	None; sublobar concordance with iEEG	Concordance 80.1 % temporal vs 92.0 % extratemporal	No counterpart for concordance. For the authors’ explanation (depth, helmet distance): within the dense array, detectability of temporal over frontal targets 0.73 (Neuromag 0.79); with a 6-mm OPM gap, detectability against Neuromag dense 1.04, site-matched 0.96 (without the gap, 1.14 and 1.01)	–	Metric (localization concordance); comparator; dual-axis sensors in a rigid helmet, modeled only as a uniform gap; extratemporal group vs the frontal lobe; ages 6–60 vs one adult; band (2–70 Hz)

511 An analogue shares at most a comparator, a kind of metric or an age range with the study, so study and model are compared in direction only (Direction: whether the analogue
 512 points the same way as the study’s reported result); the last column names what is not matched. Two differences apply to every row: the studies recorded real discharges in
 513 patients, whereas the model computes the noise-free fields of single cortical dipoles against modeled noise in anatomies of people without epilepsy; and the model’s OPMs are
 514 idealized single-axis magnetometers with white noise, and its Neuromag a VectorView geometry with T3 coil definitions, TRIUX datasheet noise and no signal-space
 515 separation. The model’s peak-channel SNR divides by the modeled noise only, the studies’ by a baseline that also holds physiological noise. Unless stated, model values are
 516 medians over cortical targets with sensor plus brain noise, OPM noise of 15 fT/√Hz and a band of 1–40 Hz: OPM/Neuromag ratios for the adult at its measured head position
 517 (all 7,661 targets, unweighted, medial wall included; brackets, 95 % parcel-bootstrap intervals) and, for the school-age scaled adult, *D* in dB at top contact in the fixed adult

518 helmet (area-weighted, medial wall excluded). Site-matched: the site-matched OPM array (98 sites), the array closest in channel count to the larger clinical arrays; dense: 208
519 sites. Literature values are as reported. ^dDerived from printed values. ^pValue or detail seen only in the study's preprint (for Feys et al. (2022), whose published version was read
520 as its abstract together with the full preprint). ^aValues from the abstract only (literature search and extraction: Section S8.3). Rb, rubidium vapor sensors; ⁴He, helium-4
521 sensors; IED, interictal epileptiform discharge; SD, standard deviation; tSSS, temporally extended signal-space separation; SEEG, stereo-electroencephalography; iEEG,
522 intracranial EEG; CTF and 4D, commercial SQUID systems.

4.4. Limitations

This study has limitations. First, no measurement validates the OPM covariance: the OPM sensor noise is declared and white, and Neuromag’s measured noise alone favors Neuromag by as much as the dense array’s whole advantage. A measured OPM noise covariance in the same band, under recording conditions matched to those of the recording that calibrates the model, would test the predicted advantage, and public OPM recordings could give a first check. Second, a single projection stands in for the preprocessing of real systems (Taulu and Simola, 2006; Tierney et al., 2021), which applying each system’s own interference suppression would test. Third, one adult is the reference (its D spans 0.46 dB over the 12 source-blind placements), and the pediatric results rest on two copies of it, three templates and three children from one dataset. The parcel intervals exclude between-subject variability, which a cohort of individual pediatric MRIs would measure. Fourth, the children’s skull is modeled and their fiducials are transferred, and the MRI check cannot verify the white surface within a few millimeters of the scalp. Nor can it detect a head boundary that the T1 images themselves place too deep, which could explain both the shallow cortex and the dataset’s own inner skull lying so close to the scalp. The children’s negative Δ at the adult’s gap may rest on that near-scalp cortex, which is why we treat them as provisional examples; individual pediatric MRIs with an independently measured scalp and skull and digitized fiducials would test it. Fifth, the templates’ scalps lie 1.3–1.6 mm outside their MRI head boundaries and the adult’s 1.3 mm inside, so sensors sit farther from a template’s brain than from the adult’s. The templates’ D and Δ , including Δ in the fitted helmet, may therefore carry a difference of conventions of 2.6–3.0 mm that could make them too low; for illustration only, a 3-mm OPM gap lowers the adult’s dense ratio from 1.14 to 1.09. The within-head contrast compares two helmets on one scalp, and the interaction is a difference of two such contrasts, so both are less exposed to it. The 18-month template’s white surface also fits its T1 image best after a rigid shift of about 2.3 mm, a possibly displaced cortex or only the weak contrast of an average image. Such a shift brings part of the cortex closer to the sensors and part farther, so even the sign of its effect on this template’s Δ is not obvious; it was not estimated. Template scalps derived from the templates’ own MRI head boundaries, and a re-registered 18-month surface, would remove both questions. Sixth, the smaller heads keep the adult’s background variance per unit cortical area. With that variance multiplied by 0.5 or 2, every smaller head’s median D still exceeded the adult’s in the fixed helmet (by +0.25 to +1.18 and +0.11 to +1.07 dB), but measured pediatric brain noise would settle it. Seventh, all sources lie on the cortical surface, so the hippocampus and amygdala are absent and would need volume sources. Finally, the spike simulations are idealized: the detector’s templates, candidate patterns and whitening covariance come from the simulation’s own stationary model, and the null data hold no physiological transients. The ranking was tested against one declared mismatch under the primary noise model only, and head motion was bounded separately rather than simulated with the spikes (Section S7). Recordings of real or phantom spikes with both systems would test the detection result.

5. Conclusions

With explicit noise in a realistic adult head, a dense array of 15-fT/ $\sqrt{\text{Hz}}$ OPMs had a modestly higher detectability than the Neuromag system, an array at Neuromag’s sites showed no established advantage, and OPM noise or array design could reverse which system led. The dense advantage is as large as the model’s error for Neuromag; its break-even OPM noise, 31.5 [28.5, 34.9] fT/ $\sqrt{\text{Hz}}$, is a prediction for measured OPM noise to test. Fitting the helmet to the adult’s gap removed most of the extra advantage that the scaled adults and templates showed in the fixed adult helmet; the children are provisional examples. In a confirmatory run (idealized detector,

modeled noise, fixed adult helmet), the dense array detected weaker superficial simulated spikes in all nine anatomies (not independent replications), without an established localization difference (exploratory). Clinical reports of higher OPM SNR are compatible with these conditions, which the model cannot tell apart in any single study.

CRediT authorship contribution statement

Laiba Khan: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Mainak Jas:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The analysis code, configurations and tests are available at <https://github.com/khan-laiba/opm-squid-meg> under the MIT license, and the result files (JSON and CSV, in the repository’s results folder) and figures under the CC BY 4.0 license. The external data are not redistributed but are obtained from their sources with the repository’s fetch scripts: the MNE sample dataset (Gramfort et al., 2013), in the processed release that the dataset fetcher of MNE-Python 1.13.2 downloads (OSF file 86qa2, version 6); the infant templates (O’Reilly et al., 2021; Richards et al., 2016), through MNE-Python’s template fetcher (LGPL-2.1); and the MRIs and surfaces of the children, OpenNeuro ds005234, version 2.2.0 (Fadeev et al., 2025), whose metadata give CC0 and whose authors ask that it be acknowledged as distributed under CC BY with its DOI. The documentation of the sample dataset restricts it to learning the software and advises against using it to evaluate the MEG or MRI system. Here, the recording calibrates the noise model and its measured covariance serves as a check of that model, and the values are not offered as an evaluation of the VectorView system or of the MRI scanner.

Ethics statement

This simulation study used only publicly available, de-identified data collected under the approvals of the original studies; no new data were collected from human participants. Sex and gender were not analyzed: with one adult, three averaged templates and three children, the head models are too few for any analysis by sex.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to write and run the analysis code and to draft and edit the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Supplementary material

Supplementary material related to this article is provided as a separate file. It comprises Sections S1–S8 (estimands and conventions; the spherical benchmark; the noise model's components, calibration, validation and sensitivity analyses; the head models and the MRI check; the helmet constructions; regions and the noise floor across heads; the spike study; and the parameters, the analysis choices made after results were known, the literature search and the parameters' provenance), Figs. S1–S15 and Tables S1–S18.